BACS1014

Problem Solving And Programming

Assignment

2023/2024

Programme	:	RDSY1S1 -Bachelor of Computer Science (Honours) in Data Science
Tutorial Group	:	Group 3
Tutor Name	:	Teh Boon Chuan
Date Submitted to Tutor	:	26/12/2023

Team Members:

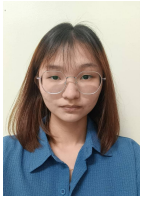
No	Student Name	Student ID
1.	Wong Jin Xuan	2314630
2.	Dorcas Lim Yuan Yao	2314535
3.	Tan Yen Ping	2314615

No.	Team Member	Task(s) Completed	Overall Contribution (%)
1.	Wong Jin Xuan	Program: Seat Display (VIP, General), Order Ticket, Payment Report: Pseudocode, Program Testing and Output	33.3
2.	Dorcas Lim Yuan Yao	Program: Receipt, Calculate, Login Report: Description/Purpose, Additional functionalities	33.3
3.	Tan Yen Ping	Program: Sales report, Ticket, Menu Report: Constant and Variables, Structure Chart	33.3

Coursework Declaration

We confirm that we have read and shall comply with all the terms and conditions of TAR University of Management and Technology plagiarism policy.

We declare that this assignment is free from all forms of plagiarism and for all intents and purposes is our own properly derived work.

Name :	Wong Jin Xuan	Dorcas Lim Yuan Yao	Tan Yen Ping
Photo :			
Signature :			

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Assessment Criteria Form

Prog/Grp: RDSY1S1 G3

Team Name: **Wong Jin Xuan, Dorcas Lim Yuan Yao, Tan Yen Ping**

Team Mark: **/80 (Full mark)**

	Student names (alphabetical order):	**Individual Mark* / 20**	**Contribution (%)**	**TOTAL MARKS**
1	Dorcas Lim Yuan Yao			**0**
2	Tan Yen Ping			**0**
3	Wong Jin Xuan			**0**

***Ability to construct a program to solve given problems by using well-structured programming design (verified via assignment contribution and Q&A during presentation)**

	Aspect	Criteria	Weight	Mark Awarded (0..5)	Weighted Mark
1	**Constants & Variables**	Proper identification and use of constants and variables	2		
2	**Scope of variables**	Use local variables and NOT global variables	2		
3	**Menu Processing**	Use of proper loop, process of menu options, handle invalid options, exit option, etc.	3		
4	**Input**	Correct and clear instructions for input and capture of input data	3		
5	**Input validation**	Provide at least 3 correct validations	2		
6	**Output**	Correct and clear output display for tickets and reports with correct use of various manipulators	3		
7	**User friendliness**	Overall screen design is clear, tidy and organised. Easy to navigate and understand the program flow.	2		
8	**Use of arrays**	Proper declaration, correct use of index, accessed & updated using for loop for seats reservation.	4		
9	**Selection control structure**	Correct & appropriate use of control structures like if else, switch, nested if & nested switch. Proper condition specified.	4		
10	**Repetitve structure**	Correct & appropriate use of looping	4		

		structures in each situation, correct syntax and positioning of initialisation, condition & updating. Correct & appropriate use of nested looping (if any).			
11	**Functions**	Correct & clear break down of functions that perform specific tasks. Correct declaration, definition & calls. Use of parameters in various passing methods & return statement.	4		
12	**Basic requirements met**	The program works as intended and all the 10 requirements are met (refer to question) ?? 10?	9		
13	**Extra features/ functionalities**	Provide 2 extra meaningful working extra features	9		
14	**Readability**	Meaningful identifiers names used. Clear indentation and program constructs. Clear & useful comments used.	4		
15	**Originality & creativity**	Uniqueness & creativity introduced as compared to the required basic layout and functionalities.	4		
16	**Report: Introduction**	Clear and useful description (one page)	2		
17	**Report: Constants & variables**	Correct list with clear description for data type, purpose, constant values.	2		
18	**Report: Test Cases**	Clear description of test scenarios, at least 3 sets, complete and clear screenshots matching the planned scenarios.	3		
19	**Report: Two extra features**	Well explained with diagrams or screenshots	4		
20	**Report: Structure and layout**	Clear table of contents, header & footer clear sections with subtitles, page numbers, consistent font type and size used, tidy, well organised, etc.	2		
21	**Program design: Structure Chart**	Correct modules & flow, At least 3 levels, Clear and tidy	4		
22	**Program design: Flowchart/Pseudocode**	Correct symbols used, Clear labels, Indentation and tidy	4		

Brief Description / Purpose of the Program

The purpose of the Concert Ticketing System is to support the concert organiser in efficiently managing ticket orders for the upcoming Cool Play Concert, providing a streamlined and user-friendly ordering process implemented through a C++ program. As the concert is expecting a large crowd of audience, this program plays a crucial role in assisting organisers to sell, process, track, and handle tickets with the ticketing system as efficiently as possible. The system includes features such as displaying a ticket ordering menu, presenting seat layouts with pricing, prompting users for ticket preferences, calculating total payments, printing receipts, accepting various payment methods, generating and displaying tickets, and producing a daily sales report.

Purpose:

1. **Efficient Ticket Ordering**
   The system offers users a user-friendly menu that ensures a seamless ticket-ordering experience. Fans of the Cool Play Concert are able to view all the ticket categories, pick their preferred ones, and indicate how many tickets they want.

2. **Seamless Seat Reservation**
   Audiences of the concert can view the layout and price of seats that are available for each ticket category, which helps them in making decisions and booking the seats that they want. With the help of this function, users can have a more streamlined and delightful ticket selection procedure overall.

3. **Payment Processing**
   Based on the choices made by the user, the system calculates the total amount charged and provides several payment options. The system offers clients the right to facilitate payments through several channels, including cash, cards, and e-wallets, while ensuring a safe and easy transaction process.

4. **Ticket Generation and Display**
   Upon a successful payment, the system generates ticket(s) and receipt (optional) that are well-formatted and contain accurate information. Customers are then given these

ticket(s) as a confirmation of their purchase. This feature makes the ticketing system more practical and professional.

5. **Sales Reporting**

   The system constantly tracks daily ticket sales to assist with event management. Through the system's insights regarding profit and ticket sales, event organisers are able to make appropriate choices and enhance concert planning.

The Concert Ticketing System implements the guidelines for C++ development. It includes fundamental ideas and structures, like arrays for seat arrangements, loops for managing menus and orders, and proper use of variables and constants to improve the understanding and maintainability of the code.

Overall Objective:

Providing a complete, effective, and user-friendly ticket sales management system is the main goal of the Concert Ticketing System. Through a combination of most effective programming practises, thoughtful options, and the special feature of login function and printing receipts, the system aims to offer concertgoers a pleasurable and memorable experience, while at the same time enabling organisers to organise concerts efficiently.

Constants and Variables

Constants

Identifier	Data type	Value	Description / Purpose
RED_TEXT	string	“\033[1;31m”	Set the text colour to red.
TOTAL_TICKETS	integer	170	Defines the total number of available tickets for sale.
ticketType	array of string	{“VIP”, “General”}	Stores ticket types which are VIP and GENERAL.
ticketPrice	array of double	{788.90, 259.90}	Stores ticket price for different categories
VIPROW	integer	3	Appear for the number of rows in the VIP seat section.
VIPCOL	integer	10	Appear for the number of columns in the VIP seat section.
GENROW	integer	7	Appear for the number of rows in the General seat section.
GENCOL	integer	20	Appear for the number of columns in the General seat section.

Variables

Global variable			
Identifier	**Data type**	**Sample value**	**Description / Purpose**
tickets_sold	integer	120, 34 (max 170)	Used to track the total number of tickets sold that day.
cust_num	integer	3, 42	Used to track the number of customers.
vipAvailib	integer	30, 21	Used to track the number of available VIP seats.
generalAvailib	integer	140, 96	Used to track the number of available general seats.

currentVIPAmount	integer	24, 17	Stores the current amount (quantity) of VIP tickets being processed.
currentGenAmount	integer	77, 120	Stores the current amount (quantity) of general tickets being processed.
currentVIPSeats	2-dimensional array of boolean	{{true, false, true … }, {false, …}, …}	Used to track the current VIP seats chosen by the user. True indicates the seats chosen, otherwise False.
currentGenSeats	2-dimensional array of boolean	{{true, false, true … }, {false, …}, …}	Used to track the current general seats chosen by the user. True indicates the seats chosen, otherwise False.
currentTotal	double	788.90, 1039.80	Stores the current total of payment by the customer.
vipSeatsReserv	2-dimensional array of boolean	{{true, false, true … }, {false, …}, …}	Used to store seats reserved and sold out for VIP categories. True indicates the seats reserved, otherwise False.
genSeatsReserv	2-dimensional array of boolean	{{true, false, true … }, {false, …}, …}	Used to store seats reserved and sold out for General categories. True indicates the seats reserved, otherwise False.

Local variable				
Module / function	**Identifier**	**Data type**	**Sample value**	**Description / Purpose**
main( )	onGoing	boolean	true, false	Used as an event controller in a do-while loop. True if the user continues to use the program. False if the user chooses to quit the program. Variable passed by address into menu function.
login( )	username	string	“user1”	Used to store the correct username.
	password	string	“password”	Used to store the correct password.
	usernameInput	string	“user1”, “UserOne”	Stores the username input by the user.

	pswInput	string	“password”, “PAss Wod”	Stores the password input by the user.
menu( )	menuOpt	character	‘1’, ‘2’, ‘3’, ‘4’	Stores the user’s input for menu option choice.
	*onGoing	boolean pointer	address of variable onGoing in main()	Change the value to false when the user chooses menu option [4].
seatDisplay( )	category	character	‘1’, ‘2’, ‘3’	Stores the seat category choice selected by the user.
	action	character	‘1’,’2’	Stores the user’s choice for further action after displaying seats.
orderTicket( )	category	character	‘1’, ‘2’	Stores the seat category choice selected by the user.
	confirm	character	‘Y’, ‘y’, ‘N’, ‘n’	Used to accept user confirmation for payment after calculating the total price to pay. ‘Y’ for agreeing to pay, ‘N’ for cancel ordering.
	repeat	character	‘Y’, ‘y’, ‘N’, ‘n’	Used to accept user input regarding whether they would like to order additional tickets. ‘Y’ to repeat adding and ordering tickets, ‘N’ to not continue adding tickets.
	back	character	‘Y’, ‘y’	Used in the case when all tickets for the chosen category had sold out. Only enter ‘Y’ to go back to the menu.
	seatsQuantity	integer	3, 1, 5	Used to store the current quantity of seats chosen. Variable passed by address into validateAmount and validateSeats functions.
	payMethod	string	“Cash” , “E-Wallet”	Stores the payment method for the current transaction. Variable passed by address into payment and receipt functions.
	receiptMsg	string	“Balance :	Used to add the additional

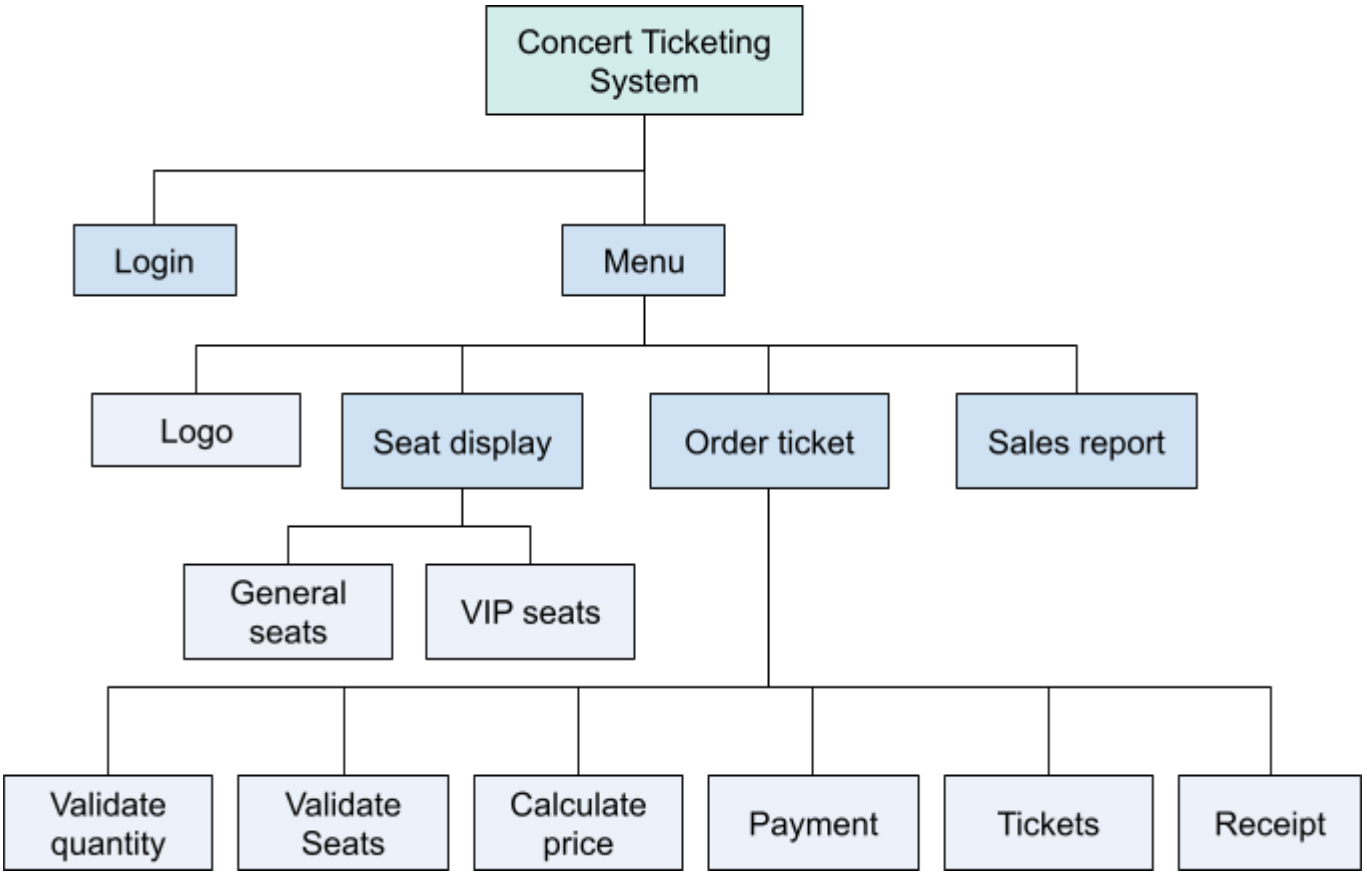
			RM211.10”, “Credit Card Ref: 1234567890123456”	information for the receipt. Variable passed by address into payment and receipt functions.
validateQuantity()	amount	integer	3, 67	Stores the amount of tickets that users want to buy.
	ticketAvailib	integer	120, 12	Assigned with global variable generalAvailib or vipAvailib based on the category chosen by the customer.
	*seatsQuantity	integer pointer	address of variable seatsQuantity in orderTicket()	Used to store the validated amount of tickets for the use in validateSeats function.
validateSeats( )	chRow	character	‘A’, ‘C’, ‘g’, ‘f’...	Used to store the seat row entered by the user in character. Seat code format: alphabet + number (e.g. A12, B3, G20)
	seatRow	integer	1, 6	Used to store converted row alphabet into integer value for easier seats array handling.
	seatCol	integer	3, 7, 20	Used to store column numbers entered by the user.
	*seatsQuantity	integer pointer	address of variable seatsQuantity in orderTicket()	Prompt the number of seat codes input based on the seat quantity chosen.
calculate( )	total	double	16028.90, 1577.80	Used to store the calculated total amount for payment.
payment( )	method	character	‘1, ‘2’, ‘3’	Stores the user’s input for payment method choice.
	creditNum	c-string	1234567890123456 (16 digit validation)	Stores the credit card number entered by the user as characters for validation.
	mm	integer	1 – 12	Store the month of the

				expiration date for credit/debit card.
	yy	integer	24 – 28	Store the year of the expiration date for credit/debit card.
	name	string	“Lee Jun Kit”, “amy lim”	Store the name on the card.
	cashPaid	double	3000, 2400	Stores the amount of cash paid by the user.
	integerPart	integer	1600, 5000	Store the integer part of the balance for cash paid for data type handling.
	*receiptMsg	string pointer	address of variable receiptMsg in orderTicket()	Store extra message for receipt printing based on the payment method chosen
	*payMethod	string pointer	address of variable receiptMsg in orderTicket()	Store the payment method for receipt printing
receipt( )	confirm	character	‘Y’, ‘y’, ‘N’, ‘n’	Used to accept user confirmation for printing receipt after payment. ‘Y’ for generating receipt, ‘N’ for no receipt.
	seatRow	string	“A”, “VB”	Used to store the seat row alphabet for receipt printing.
	seatCol	string	“6”, “20”	Used to store the seat column number for receipt printing.
	*receiptMsg	string pointer	address of variable receiptMsg in orderTicket()	Print extra message for receipt printing based on the payment method chosen.
	*payMethod	string pointer	address of variable receiptMsg in orderTicket()	Print the payment method for receipt printing
ticket( )	seatRow	string	“A”, “VB”	Used to convert the array number for seats chosen into a string that stores the row

				alphabet.
	seatCol	string	“6”, “20”	Used to convert the array number for seats chosen into a string that stores the column number.
	next	character	‘Y’, ‘y’	Used to accept user confirmation to loop back the menu after done section. Only enter ‘Y’ to go back.
	queue	integer	15, 140	Used to store the queue number based on the sequence of tickets sold.
salesReport( )	next	character	‘Y’, ‘y’	Used to accept user confirmation to back to the menu. Only enter ‘Y’ to go back.
	vipSold	integer	3, 23	Used to store the number of VIP seats sold.
	genSold	integer	44, 136	Used to store the number of general seats sold.
	vipProfit	double	3944.50, 1577.80	Used for storing the profit generated from the sale of VIP tickets.
	genProfit	double	1003.60, 501.80	Used for storing the profit generated from the sale of general tickets.
	data	array of integer	{ 30, 120 }	Used to store data for the number of dots to be displayed in the bar chart.
	higher	string	“VIP”, “General”	Used to store the category with higher data (tickets sold).
	lower	string	“VIP”, “General”	Used to store the category with lower data (tickets sold).

Overall Program Design

Structure Chart



Pseudocode

```
START
   1. Initialize boolean variable onGoing to true
   2. Call login()
   3. Do While tickets_sold < TOTAL_TICKETS and onGoing (true)
   4.       Set currentVIPAmount = 0
   5.       Set currentGenAmount = 0
   6.       Set currentTotal = 0
   7.       For row = 0; row < VIPROW; row++
   8.            For col = 0; col < VIPCOL; col++
   9.                 currentVIPSeats[row][col] = false
   10.           End for
   11.      End for
   12.      For row = 0; row < GENROW; row++
   13.           For col = 0; col < GENCOL; col++
   14.                currentGenSeats[row][col] = false
   15.           End for
   16.      End for
   17.      Call menu(&onGoing)
   18.   End Do
   19.   Display "Thank you and have a nice day"
END

FUNCTION void login()
   1. Call printTitle()
   2. Declare and Initialize username, password, usernameInput,
      pswInput
   3. Display login header
   4. Do While username != usernameInput
   5.       Get usernameInput
   6.       If username != usernameInput then
   7.            Display error message
   8.       End If
   9. End Do
   10.   Do While (pswInput != password)
   11.      Get pswInput
   12.      If pswInput != password then
   13.           Display error message
   14.      End If
   15.   End Do
END FUNCTION

FUNCTION void printTitle()
   1. Display Cool Play Concert Ticketing System ASCII art
END FUNCTION

FUNCTION void menu(bool *onGoing)
```

```
   1. Call printTitle()
   2. Display menu header
   3. Display menu options
   4. Get menuOpt
   5. While menuOpt != 1, 2, 3, 4
   6.       Display error message
   7.       Get menuOpt
   8. End While
   9. Switch menuOpt
   10.   Case '1': Call seatDisplay()
   11.   Case '2': Call orderTicket()
   12.   Case '3': Call salesReport()
   13.   Case '4': onGoing = false
   14.   End Switch
END FUNCTION

FUNCTION void seatDisplay()
   1. Display Guitar ASCII art
   2. Display seats display header
   3. Display category option
   4. Get category option
   5. While category != '1' and category != '2' and category != '3'
   6.       Display error message
   7.       Get category option
   8. End while
   9. If category == 1 then
   10.      Display General Seats header
   11.      Call generalSeats()
   12.   Else if category == 2 then
   13.      Display VIP Seats header
   14.      Call vipSeats()
   15.   Else
   16.      Display stage layout
   17.      Call vipSeats()
   18.      Call generalSeats()
   19.   End If
   20.   Display action option
   21.   Get action option
   22.   While action != '1' and action != '2'
   23.      Display error message
   24.      Get action option
   25.   End While
   26.   If action == '2' then
   27.      Call orderTicket()
   28.   End If
END FUNCTION

FUNCTION void generalSeats()
```

```
   1. For i = 0; i < GENROW; ++i
   2.      For j = 1; j <= GENCOL; ++j
   3.           If genSeatsReserv[i][j - 1](true) then
   4.                Display seats in red
   5.           Else
   6.                Display Seats
   7.           End If
   8.           If j == 5 or j == 15 then
   9.                Display blank space
   10.          End If
   11.     End For
   12.     Display endl
   13.  End For
END FUNCTION

FUNCTION void vipSeats()
   1. For i = 0; i < VIPROW; ++i
   2.      For j = 1; j <= VIPCOL; ++j
   3.           If vipSeatsReserv[i][j - 1](true)
   4.                Display seats in red
   5.           Else
   6.                Display Seats
   7.           End If
   8.      End For
   9.      Display endl
   10.  End For
END FUNCTION

FUNCTION void orderTicket()
   1. Do While repeat == 'Y'
   2.      Display order ticket header
   3.      Display ticket table
   4.      For i = 0; i < 2; ++i
   5.           Display i, ticketType[i], ticketPrice[i]
   6.      End For
   7.      Get ticket category
   8.      While category != '1' and category != '2'
   9.           Display error message
   10.          Get ticket category
   11.     End While
   12.     If category == '1' and vipAvailib == 0 or category == '2'
      and genAvailib == 0 then
   13.          Display message stating no tickets left, buy other
      category
   14.          Get back to menu confirmation
   15.          While back != 'Y'
   16.               Display error message
   17.               Get back to menu confirmation
```

```
            End While
            Break
        End if
        If validateQuantity(category, &seatsQuantity) and
    validateAmount(category, &seatsQuantity)then
            Get add ticket confirmation (repeat)
            While repeat != 'Y' and repeat != 'N'
                  Display error message
                  Get add ticket confirmation
            End While
        End If
    End Do
    If back != 'Y' then
        currentTotal = calculate()
        Get confirmation for payment
        While confirm != 'Y' and confirm != 'N'
            Display error message
            Get confirmation
        End While
        If confirm == 'Y' then
            Call payment(currentTotal, &receiptMsg, &payMethod)
            Call receipt(&receiptMsg, &payMethod)
            tickets_sold += currentVipAmount
            Tickets_sold += currentGenAmount
            cust_num += 1
            Call ticket()
        Else
            For row = 0; row < VIPROW; ++row
                  For col = 0; col < VIPCOL; ++ col
                        If currentVIPSeats[row][col] (true) then
                              vipSeatsReserv[row][col] = false
                        End If
                  End For
            End For
            For row = 0; row < GENROW; ++row
                  For col = 0; col < GENCOL; ++ col
                        If currentGenSeats[row][col] (true) then
                              genSeatsReserv[row][col] = false
                        End If
                  End For
            End For
            vipAvailib += currentVIPAmount
            generalAvailib += currentGenAmount
        End If
    End If
END FUNCTION

FUNCTION bool validateQuantity(char category, int *seatsQuantity)
```

```
   1. If category == '1' then
   2.       ticketAvailib = vipAvailib
   3.  Else
   4.       ticketAvailib = generalAvailib
   5. End If
   6. Do While amount > ticketAvailib or amount <= 0
   7.       Get amount of ticket to buy
   8.       If amount > ticketAvailib then
   9.            Display error message, amount of seats left
   10.      Else if amount <= 0 then
   11.           Display error message
   12.      End If
   13.   End Do
   14.   If category == '1' then
   15.      currentVIPAmount += amount
   16.      vipAvailib -= amount
   17.   Else
   18.      currentGenAmount += amount
   19.      generalAvailib -= amount
   20.   End If
   21.   *seatsQuantity = amount
   22.   Return true
END FUNCTION

FUNCTION bool validateSeats(char category, int *seatsQuantity)
   1. If category == '1' then
   2.       Call vipSeats()
   3. Else
   4.       Call generalSeats()
   5. End If
   6. For i = 1; i <= *seatsQuantity; i++
   7.       Get seat code (chRow, seatCol)
   8.       seatRow = chRow - 'A'
   9.       If category == '1' then
   10.           While seatRow < 0 or seatRow >= VIPROW or
                 seatCol <= 0 or seatCol > VIPCOL or
                 vipSeatsReserv[seatRow][seatCol - 1] (true)
   11.                If vipSeatsReserv[seatRow][seatCol - 1] (true)
      then
   12.                     Display seat not available
   13.                Else
   14.                     Display invalid code
   15.                End If
   16.                Get seat code (chRow, seatCol)
   17.                seatRow = chRow - 'A'
   18.           End While
   19.           vipSeatsReserv[seatRow][seatCol - 1] = true
   20.           currentVIPSeats[seatRow][seatCol - 1] = true
```

```
   21.     Else
   22.          While seatRow < 0 or seatRow >= GENROW or
                seatCol <= 0 or seatCol > GENCOL or
                genSeatsReserv[seatRow][seatCol - 1] (true)
   23.               If genSeatsReserv (true) then
   24.                    Display seat not available
   25.               Else
   26.                    Display invalid code
   27.               End If
   28.               Get seat code (chRow, seatCol)
   29.               seatRow = chRow - 'A'
   30.          End While
   31.          genSeatsReserv[seatRow][seatCol - 1] = true
   32.          currentGenSeats[seatRow][seatCol - 1] = true
   33.     End If
   34.  End For
   35.  *seatsQuantity = 0
   36.  Return true
END FUNCTION

FUNCTION double calculate()
   1. Display price table header
   2. If currentVIPAmount != 0 then
   3.      Display ticketType[0], ticketPrice[0], currentVIPAmount,
      currentVIPAmount * price
   4. End If
   5. If currentGenAmount != 0 then
   6.      Display ticketType[1], ticketPrice[1], currentGenAmount,
      currentGenAmount * price
   7. End If
   8. total = currentVIPAmount * ticketPrice[0] + cuurentGenAmount *
      ticketPrice[1]
   9. Display total payment
   10.  Return total
END FUNCTION

FUNCTION void payment(double total, string *receiptMsg, string
*payMethod)
   1. Display payment section header
   2. Display payment option
   3. Get payment method
   4. While method != '1' and method != '2' and method != '3'
   5.      Display error message
   6.      Get payment method
   7. End While
   8. If method == '1' then
   9.      *payMethod = "Credit / Debit Card"
   10.     Get credit card number
```

```
   11.      While strlen(creditnum) != 16
   12.           Display error message
   13.           Get creditnum
   14.      End While
   15.      Get expiry month (mm)
   16.      While mm < 1 or mm > 12
   17.           Display error message
   18.           Get expiry month
   19.      End While
   20.      Get expiry year (yy)
   21.      While yy < 24 or yy > 28
   22.           Display error message
   23.           Get expiry year
   24.      End While
   25.      Get name on card
   26.      *receiptMsg = "Card Payment Ref:" + creditnum
   27.      Display transaction successful
   28.   Else if method == '2' then
   29.      *payMethod = "Cash"
   30.      Get cash paid
   31.      While cashPaid < 0 or cashPaid < total
   32.           If cashPaid < 0 then
   33.                Display error message for positive value
   34.           Else
   35.                Display more payment needed
   36.           End If
   37.           Get cash paid
   38.      End While
   39.      If cashPaid > total then
   40.           Display balance
   41.      End If
   42.      Display payment succeed
   43.      *receiptMsg = "Balance:" + balance
   44.   Else
   45.      *payMethod = "E-Wallet"
   46.      Display transaction successful
   47.   End If
END FUNCTION

FUNCTION void receipt(string *receiptMsg, string *payMethod)
   1. Get confirmation to print receipt
   2. While confirm != 'Y' and confirm != 'N'
   3.      Display error message
   4.      Get confirmation
   5. End While
   6. If confirm == 'Y' then
   7.      Display receipt header layout
   8.      If currentVIPAmount != 0 then
```

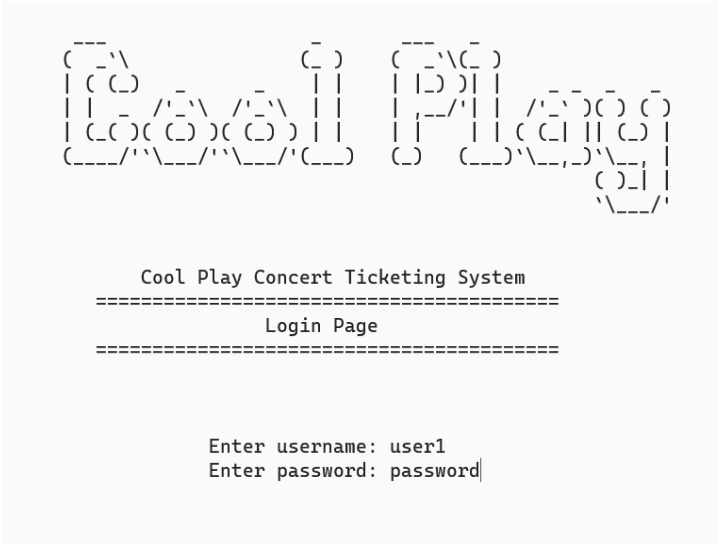
```
   9.            Display ticketType[0], currentVIPAmount,
      ticketPrice[0], currentVIPAmount * price
   10.      End If
   11.      If currentGenAmount != 0 then
   12.            Display ticketType[1], currentGenAmount,
      ticketPrice[1], currentGenAmount * price
   13.      End If
   14.      Display seats code (use for loop)
   15.      Display currentTotal, *payMethod, *receiptMsg
   16.   End If
   17.   Set *receiptMsg and *payMethod to “”
END FUNCTION

FUNCTION void ticket(int category)
   1. int queue = tickets_sold + 1
   2. Display ticket generated header
   3. For i = 0; i < VIPROW; ++i
   4.      For j = 1; j <= VIPCOL; ++j
   5.            If currentVIPSeats[i][j - 1](true) then
   6.                  seatRow = “V” + string(1, char(i + ‘A’))
   7.                  seatCol = to_string(j);
   8.                  queue -= 1
   9.                  Display VIP ticket layout
   10.           End If
   11.      End For
   12.   End For
   13.   For i = 0; i < GENROW; ++i
   14.      For j = 1; j <= GENCOL; ++j
   15.           If currentVIPSeats[i][j - 1](true) then
   16.                 seatRow = string(1, char(i + ‘A’))
   17.                 seatCol = to_string(j);
   18.                 queue -= 1
   19.                 Display general ticket layout
   20.           End If
   21.      End For
   22.   End For
   23.   Display "Press [Y] if done ... ..."
   24.   Get input to continue
   25.   While next != ‘Y’
   26.      Display error message
   27.      Get input to continue
   28.   End While
END FUNCTION

FUNCTION void salesReport()
   1. vipSold = VIPROW * VIPCOL - vipAvailib
   2. genSold = GENROW * GENCOL - generalAvailib
   3. vipProfit = vipSold * ticketPrice[0]
```

```
   4. genProfit = genSold * ticketPrice[1]
   5. Display Sales Report header
   6. Display date, event, user
   7. Display Current Sales header
   8. Display table containing price, total seats, seat left, seat
      sold, profit for both categories
   9. Display total profit and number of customers
   10.   Initialize array int data[2] = {vipSold, genSold}
   11.   Set string higher = ticketType[0]
   12.   Set string lower = ticketType[1]
   13.   If genSold > vipSold then
   14.      data[0] = genSold
   15.      data[1] = vipSold
   16.      higher = ticketType[1]
   17.      lower = ticketType[0]
   18.   End If
   19.   For int dot = data[0]; dot > 0; dot--
   20.      For int bar = 0; bar < 2; bar++
   21.           If data[bar] >= dot then
   22.                Display "*"
   23.           Else
   24.                Display " "
   25.           End If
   26.      End For
   27.      Display endl
   28.   End For
   29.   Display higher, lower
   30.   Display "Press [Y] back to menu: "
   31.   Get input to continue
   32.   While next != 'Y'
   33.      Display error message
   34.      Get input to continue
   35.   End While
END FUNCTION
```

Addition Functionalities

User Login

The login feature for staff serves as a crucial tool for access control and security. The system ensures that only authorised users can access and administer the ticketing functionalities, sales reports, and other administrative features by requiring staff members to authenticate themselves through a login process. By limiting illegal access and protecting private data, this login feature serves as the system's main gatekeeper. It gives the system an extra degree of traceability and responsibility, enabling it to monitor employee actions and keep an audit trail for security reasons. The login option guarantees a safe and regulated setting for efficient concert ticket administration.

Test data	Expected output	Actual output
Scenario 1 Correct username (user1) Correct password (password)	No error message prompt. Direct to the menu page.	

		``` Welcome to concert ticketing system for ---------- Cool Play Concert ---------- 1 -- Seat display 2 -- Order ticket 3 -- Sales report 4 -- Close system Select an option: ```
Scenario 2 Incorrect username input (e.g. user, 222)	Display error message and continue to get username until the correct username is entered.	``` Cool Play Concert Ticketing System ========== Login Page ========== Enter username: user Invalid username. Please reenter. Enter username: 222 Invalid username. Please reenter. Enter username: ```
Scenario 3 Incorrect password input (e.g. passsword, abc123)	Display error message and continue to get password until the correct password is entered.	``` Cool Play Concert Ticketing System ========== Login Page ========== Enter username: user1 Enter password: passsword Wrong password. Try again. Enter password: abc123 Wrong password. Try again. Enter password: ```

Generate Receipt

The Concert Ticketing System generates receipts for customers as a way to verify and record completed transactions and give them a physical record of their ticket purchase. Important information on the receipt is the ticket category, quantity, total amount paid, and the method of payment. Receipts help organiser to track income and ensure compliance with rules in addition to assist with sales reporting procedures. This extends beyond providing consumer assurance. This feature improves the customer experience overall by giving the ticketing process a professional feel and boosting trust in the system's accuracy and efficiency.

Test data	Expected output	Actual output
Scenario 1 Receipt generated for credit/debit card payment method	Variable for receipt is correct: Invoice number, ticket category, amount, price, seat(s) code, total price. Display the payment method chosen correctly. Show the card reference number on the receipt.	

```
 ------------------------------------------------------------
|                                                            |
|          +-----------------------------------------+        |
|          |    _____          __  ___  __           |        |
|          |   / ___/__  ___  / / / _ \/ /__ ___ __  |        |
|          |  / /__/ _ \/ _ \/ / / ___/ / _ `/ // /  |        |
|          |  \___/\___/\___/_/ /_/  /_/\_,_/\_, /   |        |
|          |                                 /___/   |        |
|          +-----------------------------------------+        |
|                      ** RECEIPT **                         |
|                                                            |
|              Invoice number  : INV0001                     |
|              Date            : 17/12/2023                  |
|   ======================================================   |
|                                                            |
|   Ticket category     Qty       Price (RM)     Amount      |
|   VIP                 2         788.90         1577.80     |
|                                                            |
|              Seat(s) code    :                             |
|                                VA1                         |
|                                VA2                         |
|                                                            |
|   ------------------------------------------------------   |
|              Total Price     : RM 1577.80                  |
|              Payment Method  : Credit / Debit Card          |
|              Card Payment Ref: 1234567890123456             |
|                                                            |
 ------------------------------------------------------------


                  Please take the receipt.
```

Scenario 2 Receipt generated for cash payment method	All information is correctly printed. Display the balance for the cash payment.	``` +----------+ | | | | ** RECEIPT ** | | Invoice number : INV0002 | | Date : 17/12/2023 | | ========== | | Ticket category Qty Price (RM) Amount | | VIP 3 788.90 2366.70 | | General 4 259.90 1039.60 | | Seat(s) code : | | VA3 | | VA4 | | VB5 | | B9 | | B10 | | B11 | | B12 | | ---------- | | Total Price : RM 3406.30 | | Payment Method : Cash | | Balance : RM 93.70 | +----------+ Please take the receipt. ```
Scenario 3 Receipt generated for e-wallet payment method.	All information is correctly printed.	``` +----------+ | | | | ** RECEIPT ** | | Invoice number : INV0003 | | Date : 17/12/2023 | | ========== | | Ticket category Qty Price (RM) Amount | | General 4 259.90 1039.60 | | Seat(s) code : | | A6 | | A7 | | A14 | | A15 | | ---------- | | Total Price : RM 1039.60 | | Payment Method : E-Wallet | +----------+ Please take the receipt. ```

Program Testing and Output

Scenario 1

Order No.	Input				Output	
	Ticket Category		Seat Code	Payment Method	Price Payable	
	VIP	General			Total for each category	Total payment
1	2	0	VIP: [A5] [A6]	Credit / Debit card	VIP: 2*788.90 = 1577.80	1577.80
2	2	5	VIP: [A3] [a4] General: [A8] [A9] [A10] [A11] [A12]	Cash	VIP: 2*788.90 = 1577.80 General: 5*259.90 = 1299.50	1577.80 + 1299.50 = 2877.30
3	6	3	VIP: [B4] [B5] [B6] [B7] (add) [C5] [C6] General: [C6] [C7] [C8]	E-wallet	VIP: 6*788.90 = 4733.40 General: 3*259.90 = 779.70	4733.40 + 779.70 = 5513.10

Actual Output

Order 1 (Test for buying one category) :

```
 ___                    _       ___    _
(  _`\                 (_ )    (  _`\(_ )
| ( (_)   _      _     | |    | |_) )| |     _ _   _    _
| |  _  /'_`\  /'_`\   | |    | ,__/'| |  /'_` )( ) ( )
| (_( )( (_) )( (_) ) | |    | |    | | ( (_| || (_) |
(____/''\___/''\___/'(___)   (_)   (___)`\__,_)`\__, |
                                                 ( )_| |
                                                 `\___/'


    Welcome to concert ticketing system for
   ----------------------------------------
                Cool Play Concert
   ----------------------------------------


               1 -- Seat display
               2 -- Order ticket
               3 -- Sales report
               4 -- Close system

              Select an option: 2
```

```
 ----------------------------------------
            Ticket ordering
 ----------------------------------------


    No.    Ticket Type          Price
 ----------------------------------------
      1            VIP         788.90
      2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 2




  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


         Please enter seat code 1: A5
         Please enter seat code 2: A6
         Do you want add ticket? (Y/N): N
```

```
 Ticket Category        Price/ticket       Quantity         Amount
*********************************************************************
             VIP             788.90             2        1577.80
---------------------------------------------------------------------
                 Total payment: RM 1577.80
---------------------------------------------------------------------


                 Confirm Payment? (Y/N): Y


     ----------------------------------------
                  Payment Section
     ----------------------------------------


               1 -- Credit/Debit Card
               2 -- Cash
               3 -- E-Wallet

       Choose a payment method: 1

       Enter credit card number (no space): 1234567890123456

       Expiry Date (MM): 12
       Expiry Date (MM/YY): 12/25

       Enter name on card: Amy Tan


          Transaction successful.


          Print receipt? (Y/N): y
```

```
 ------------------------------------------------------------
|                                                            |
|          +-----------------------------------------+        |
|          |    _____          __  ___  __           |        |
|          |   / ___/__  ___  / / / _ \/ /__ ___ __  |        |
|          |  / /__/ _ \/ _ \/ / / ___/ / _ `/ // /  |        |
|          |  \___/\___/\___/_/ /_/  /_/\_,_/\_, /   |        |
|          |                                /___/    |        |
|          +-----------------------------------------+        |
|                       ** RECEIPT **                        |
|                                                            |
|              Invoice number  : INV0001                     |
|              Date            : 17/12/2023                  |
|   ========================================================  |
|                                                            |
|   Ticket category     Qty       Price (RM)     Amount      |
|   VIP                 2         788.90         1577.80     |
|                                                            |
|              Seat(s) code    :                             |
|                                VA5                         |
|                                VA6                         |
|                                                            |
|   ---------------------------------------------------------  |
|              Total Price     : RM 1577.80                  |
|              Payment Method  : Credit / Debit Card          |
|              Card Payment Ref: 1234567890123456             |
|                                                            |
 ------------------------------------------------------------


                 Please take the receipt.
```

```
                                  ----------------------------------------
                                            Tickets Generated
                                  ----------------------------------------



 ------------------------------------------------------------------------------------------------------------------------
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                                            | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                          |
|                      |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                        | National Stadium         |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil              |
| **FREE GLOW STICK    |   \__ \  __ / __ | _|     /  _ | \__ \                                                | Kuala Lumpur             |
|                      |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                |                          |
| GATE  15A            |                                                                                            |                          |
| SEAT  VA5            |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday                |
|                      |                      _____                                  _____                          | December 22, 2023        |
|                      |                    ( ___ )----------------------( ___ )                                  |                          |
| GATE Open 6.30pm     |                     |   |  _   _   ___   ____   |   |                                    |   ------------------   |
|                      |                     |   | \ ( ) / )_ _( )  _)\  |   |                                    |  | " All MUSIC      |  |
|                      |                     |   |  )\_/(  _| |_ | '__/  |   |                                    |  |   IS BEAUTIFUL " |  |
| Queue No. 2          |                     |   |   \_/  )_____()_(     |   |                                    |   ------------------   |
|                      |                     |___|                       |___|                                    |                          |
|                      |                    (_____)----------------------(_____)                                  | C  O  O  L  P  L  A  Y   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------------------


 ------------------------------------------------------------------------------------------------------------------------
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                                            | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                          |
|                      |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                        | National Stadium         |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil              |
| **FREE GLOW STICK    |   \__ \  __ / __ | _|     /  _ | \__ \                                                | Kuala Lumpur             |
|                      |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                |                          |
| GATE  15A            |                                                                                            |                          |
| SEAT  VA6            |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday                |
|                      |                      _____                                  _____                          | December 22, 2023        |
|                      |                    ( ___ )----------------------( ___ )                                  |                          |
| GATE Open 6.30pm     |                     |   |  _   _   ___   ____   |   |                                    |   ------------------   |
|                      |                     |   | \ ( ) / )_ _( )  _)\  |   |                                    |  | " All MUSIC      |  |
|                      |                     |   |  )\_/(  _| |_ | '__/  |   |                                    |  |   IS BEAUTIFUL " |  |
| Queue No. 1          |                     |   |   \_/  )_____()_(     |   |                                    |   ------------------   |
|                      |                     |___|                       |___|                                    |                          |
|                      |                    (_____)----------------------(_____)                                  | C  O  O  L  P  L  A  Y   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------------------


                          Press [Y] if done ... ...
```

Order 2 (Test for buying two category):

```
----------------------------------------
            Ticket ordering
----------------------------------------


   No.    Ticket Type          Price
----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 2



  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


         Please enter seat code 1: A3
         Please enter seat code 2: a4
         Do you want add ticket? (Y/N): Y
```

```
                        ----------------------------------------
                                    Ticket ordering
                        ----------------------------------------


                           No.    Ticket Type          Price
                        ----------------------------------------
                             1            VIP         788.90
                             2        General         259.90


                       Please enter the category of ticket: 2


                       Please enter amount of ticket to buy: 5



  [A1][A2][A3][A4][A5]   [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15]   [A16][A17][A18][A19][A20]
  [B1][B2][B3][B4][B5]   [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15]   [B16][B17][B18][B19][B20]
  [C1][C2][C3][C4][C5]   [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15]   [C16][C17][C18][C19][C20]
  [D1][D2][D3][D4][D5]   [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15]   [D16][D17][D18][D19][D20]
  [E1][E2][E3][E4][E5]   [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15]   [E16][E17][E18][E19][E20]
  [F1][F2][F3][F4][F5]   [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15]   [F16][F17][F18][F19][F20]
  [G1][G2][G3][G4][G5]   [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15]   [G16][G17][G18][G19][G20]


                                Please enter seat code 1: A8
                                Please enter seat code 2: A9
                                Please enter seat code 3: A10
                                Please enter seat code 4: A11
                                Please enter seat code 5: A12
                                Do you want add ticket? (Y/N): N
```

```
  Ticket Category        Price/ticket       Quantity         Amount
*********************************************************************
             VIP             788.90              2        1577.80
         General             259.90              5        1299.50
---------------------------------------------------------------------
                  Total payment: RM 2877.30
---------------------------------------------------------------------


                  Confirm Payment? (Y/N): Y


      ----------------------------------------
                   Payment Section
      ----------------------------------------


                1 -- Credit/Debit Card
                2 -- Cash
                3 -- E-Wallet

        Choose a payment method: 2
           Enter cash paid (RM): 3000.00
               Balance: RM 122.70


           Payment succeed.


           Print receipt? (Y/N): Y
```

//Balance: 3000.00 - 2877.30 = 122.70

```
 ---------------------------------------------------------------
|                                                               |
|           +----------------------------------------+          |
|           |    _____          __  ___  __          |          |
|           |   / ___/__  ___  / / / _ \/ /__ ___ __  |          |
|           |  / /__/ _ \/ _ \/ / / ___/ / _ `/ // /  |          |
|           |  \___/\___/\___/_/ /_/  /_/\_,_/\_, /   |          |
|           |                                /___/    |          |
|           +----------------------------------------+          |
|                        ** RECEIPT **                          |
|                                                               |
|               Invoice number  : INV0001                       |
|               Date            : 17/12/2023                    |
|   ========================================================    |
|                                                               |
|   Ticket category     Qty       Price (RM)     Amount         |
|   VIP                 2         788.90         1577.80        |
|   General             5         259.90         1299.50        |
|                                                               |
|               Seat(s) code    :                               |
|                                 VA3                           |
|                                 VA4                           |
|                                 A8                            |
|                                 A9                            |
|                                 A10                           |
|                                 A11                           |
|                                 A12                           |
|                                                               |
|   ---------------------------------------------------------   |
|               Total Price     : RM 2877.30                    |
|               Payment Method  : Cash                          |
|               Balance : RM 122.70                             |
|                                                               |
 ---------------------------------------------------------------


                  Please take the receipt.
```

```
                                        ----------------------------------------
                                                  Tickets Generated
                                        ----------------------------------------


 ------------------------------------------------------------------------------------------------------------------------
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            | Venue :                |
| RM    788.90           |     \  |  |  |   __| _ _|   __|          _|    |    |                                        |                        |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                          | National Stadium       |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                          | Bukit Jalil            |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                   | Kuala Lumpur           |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                    |                        |
| GATE  15A              |                                                                                            |                        |
| SEAT  VA3              |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday              |
|                        |                      _____                                  _____                             | December 22, 2023      |
|                        |                    ( ___ )----------------------( ___ )                                   |                        |
| GATE Open 6.30pm       |                     |   |  _   _   ___   ____   |   |                                        |   ------------------   |
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                                        |  | " All MUSIC      |  |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                                        |  |   IS BEAUTIFUL " |  |
| Queue No. 9            |                     |   |   \_/  )_____()_(     |   |                                        |   ------------------   |
|                        |                     |___|                        |___|                                        |                        |
|                        |                    (_____)----------------------(____)                                     | C  O  O  L  P  L  A  Y |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------------------


 ------------------------------------------------------------------------------------------------------------------------
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            | Venue :                |
| RM    788.90           |     \  |  |  |   __| _ _|   __|          _|    |    |                                        |                        |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                          | National Stadium       |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                          | Bukit Jalil            |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                   | Kuala Lumpur           |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                    |                        |
| GATE  15A              |                                                                                            |                        |
| SEAT  VA4              |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday              |
|                        |                      _____                                  _____                             | December 22, 2023      |
|                        |                    ( ___ )----------------------( ___ )                                   |                        |
| GATE Open 6.30pm       |                     |   |  _   _   ___   ____   |   |                                        |   ------------------   |
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                                        |  | " All MUSIC      |  |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                                        |  |   IS BEAUTIFUL " |  |
| Queue No. 8            |                     |   |   \_/  )_____()_(     |   |                                        |   ------------------   |
|                        |                     |___|                        |___|                                        |                        |
|                        |                    (_____)----------------------(____)                                     | C  O  O  L  P  L  A  Y |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------------------


 ------------------------------------------------------------------------------------------------------------------------
| General                |                          **COOLPLAY WORLD TOUR**                                           | Venue :                |
| RM    259.90           |    __  __              _                  __   _    _                                      |                        |
|                        |   |  \/  |            (_)               / _| | |  | |                                        | National Stadium       |
|                        |   | \  / | _   _  ___  _   ___    ___  | |_  | |_ | |__    ___                                | Bukit Jalil            |
|                        |   | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                          | Kuala Lumpur           |
|                        |   | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/                          |                        |
| GATE  15A              |   |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|                          |                        |
| SEAT  A8               |                                                                                            | Wednesday              |
|                        |             _____        _                                                                 | December 22, 2023      |
|                        |            / ____|      | |                                                                |                        |
| GATE Open 7.00pm       |           | (___   _ __  | |__    ___  _ __  ___  ___                                  |   ------------------   |
|                        |            \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|                                  |  | " WHERE MEMORIES |  |
|                        |            ____) || |_) || | | ||  __/| |  |  __/\__ \                                  |  |    ARE CREATED " |  |
| Queue No. 7            |           |_____/ | .__/ |_| |_| \___||_|   \___||___/                                  |   ------------------   |
|                        |                 | |                                                                        |                        |
|                        |                 |_|                                                                        | C  O  O  L  P  L  A  Y |
 ------------------------------------------------------------------------------------------------------------------------


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| General                |                          **COOLPLAY WORLD TOUR**                                           | Venue :                |
| RM    259.90           |    __  __              _                  __   _    _                                      |                        |
|                        |   |  \/  |            (_)               / _| | |  | |                                        | National Stadium       |
|                        |   | \  / | _   _  ___  _   ___    ___  | |_  | |_ | |__    ___                                | Bukit Jalil            |
|                        |   | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                          | Kuala Lumpur           |
```

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|                      |   | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/      |                        |
| GATE  15A            |   |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|       |                        |
| SEAT  A9             |                                                                        | Wednesday              |
|                      |             _____        _                                             | December 22, 2023      |
|                      |            / ____|       | |                                         |                        |
| GATE Open 7.00pm     |           | (___   _ __  | |__    ___  _ __  ___  ___               |   _________________    |
|                      |            \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|              |  | " WHERE MEMORIES |  |
|                      |            ____) || |_) || | | ||  __/| |   |  __/\__ \              |  |    ARE CREATED " |  |
| Queue No. 6          |           |_____/ | .__/ |_| |_| \___||_|   \___||___/               |   -----------------    |
|                      |                  | |                                                |                        |
|                      |                  |_|                                                | C  O  O  L  P  L  A  Y |
```

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| General              |                    **COOLPLAY WORLD TOUR**                          | Venue :                |
| RM    259.90         |   __  __                _                  __   _    _              |                        |
|                      |  |  \/  |              (_)                / _| | |  | |             | National Stadium       |
|                      |  | \  / | _   _  ___  _   ___    ___   | |_  | |_ | |__    ___    | Bukit Jalil            |
|                      |  | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \   | Kuala Lumpur           |
|                      |  | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/    |                        |
| GATE  15A            |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|     |                        |
| SEAT  A10            |                                                                        | Wednesday              |
|                      |             _____        _                                             | December 22, 2023      |
|                      |            / ____|       | |                                         |                        |
| GATE Open 7.00pm     |           | (___   _ __  | |__    ___  _ __  ___  ___               |   _________________    |
|                      |            \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|              |  | " WHERE MEMORIES |  |
|                      |            ____) || |_) || | | ||  __/| |   |  __/\__ \              |  |    ARE CREATED " |  |
| Queue No. 5          |           |_____/ | .__/ |_| |_| \___||_|   \___||___/               |   -----------------    |
|                      |                  | |                                                |                        |
|                      |                  |_|                                                | C  O  O  L  P  L  A  Y |
```

```
| General              |                    **COOLPLAY WORLD TOUR**                          | Venue :                |
| RM    259.90         |   __  __                _                  __   _    _              |                        |
|                      |  |  \/  |              (_)                / _| | |  | |             | National Stadium       |
|                      |  | \  / | _   _  ___  _   ___    ___   | |_  | |_ | |__    ___    | Bukit Jalil            |
|                      |  | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \   | Kuala Lumpur           |
|                      |  | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/    |                        |
| GATE  15A            |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|     |                        |
| SEAT  A11            |                                                                        | Wednesday              |
|                      |             _____        _                                             | December 22, 2023      |
|                      |            / ____|       | |                                         |                        |
| GATE Open 7.00pm     |           | (___   _ __  | |__    ___  _ __  ___  ___               |   _________________    |
|                      |            \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|              |  | " WHERE MEMORIES |  |
|                      |            ____) || |_) || | | ||  __/| |   |  __/\__ \              |  |    ARE CREATED " |  |
| Queue No. 4          |           |_____/ | .__/ |_| |_| \___||_|   \___||___/               |   -----------------    |
|                      |                  | |                                                |                        |
|                      |                  |_|                                                | C  O  O  L  P  L  A  Y |
```

```
| General              |                    **COOLPLAY WORLD TOUR**                          | Venue :                |
| RM    259.90         |   __  __                _                  __   _    _              |                        |
|                      |  |  \/  |              (_)                / _| | |  | |             | National Stadium       |
|                      |  | \  / | _   _  ___  _   ___    ___   | |_  | |_ | |__    ___    | Bukit Jalil            |
|                      |  | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \   | Kuala Lumpur           |
|                      |  | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/    |                        |
| GATE  15A            |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|     |                        |
| SEAT  A12            |                                                                        | Wednesday              |
|                      |             _____        _                                             | December 22, 2023      |
|                      |            / ____|       | |                                         |                        |
| GATE Open 7.00pm     |           | (___   _ __  | |__    ___  _ __  ___  ___               |   _________________    |
|                      |            \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|              |  | " WHERE MEMORIES |  |
|                      |            ____) || |_) || | | ||  __/| |   |  __/\__ \              |  |    ARE CREATED " |  |
| Queue No. 3          |           |_____/ | .__/ |_| |_| \___||_|   \___||___/               |   -----------------    |
|                      |                  | |                                                |                        |
|                      |                  |_|                                                | C  O  O  L  P  L  A  Y |
```

```
                                Press [Y] if done ... ...Y
```

Order 3 (Test for buying again the same category):

```
----------------------------------------
            Ticket ordering
----------------------------------------


   No.    Ticket Type          Price
----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 4



  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


         Please enter seat code 1: B4
         Please enter seat code 2: B5
         Please enter seat code 3: B6
         Please enter seat code 4: B7
         Do you want add ticket? (Y/N): Y
```

```
----------------------------------------
            Ticket ordering
----------------------------------------


   No.    Ticket Type          Price
----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 2



  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


         Please enter seat code 1: C5
         Please enter seat code 2: C6
         Do you want add ticket? (Y/N): Y
```

```
----------------------------------------
            Ticket ordering
----------------------------------------


   No.    Ticket Type          Price
----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 2


Please enter amount of ticket to buy: 3



[A1][A2][A3][A4][A5]   [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15]   [A16][A17][A18][A19][A20]
[B1][B2][B3][B4][B5]   [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15]   [B16][B17][B18][B19][B20]
[C1][C2][C3][C4][C5]   [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15]   [C16][C17][C18][C19][C20]
[D1][D2][D3][D4][D5]   [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15]   [D16][D17][D18][D19][D20]
[E1][E2][E3][E4][E5]   [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15]   [E16][E17][E18][E19][E20]
[F1][F2][F3][F4][F5]   [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15]   [F16][F17][F18][F19][F20]
[G1][G2][G3][G4][G5]   [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15]   [G16][G17][G18][G19][G20]


Please enter seat code 1: C6
Please enter seat code 2: C7
Please enter seat code 3: C8
Do you want add ticket? (Y/N): N
```

```
 Ticket Category        Price/ticket       Quantity         Amount
*********************************************************************
            VIP              788.90              6        4733.40
        General              259.90              3         779.70
---------------------------------------------------------------------
                Total payment: RM 5513.10
---------------------------------------------------------------------


                Confirm Payment? (Y/N): Y


     ----------------------------------------
                 Payment Section
     ----------------------------------------


               1 -- Credit/Debit Card
               2 -- Cash
               3 -- E-Wallet

       Choose a payment method: 3


         Transaction successful.


          Print receipt? (Y/N): Y
```

```
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|                                                               |
|            +-----------------------------------------+        |
|            |    _____          __  ___  __           |        |
|            |   / ___/__  ___  / / / _ \/ /__ ___ __  |        |
|            |  / /__/ _ \/ _ \/ / / ___/ / _ `/ // /  |        |
|            |  \___/\___/\___/_/ /_/  /_/\_,_/\_, /   |        |
|            |                                /___/    |        |
|            +-----------------------------------------+        |
|                        ** RECEIPT **                          |
|                                                               |
|                Invoice number  : INV0003                      |
|                Date            : 17/12/2023                   |
|    ==========================================================  |
|                                                               |
|    Ticket category     Qty       Price (RM)     Amount        |
|    VIP                 6         788.90         4733.40       |
|    General             3         259.90         779.70        |
|                                                               |
|                Seat(s) code    :                              |
|                                 VB4                           |
|                                 VB5                           |
|                                 VB6                           |
|                                 VB7                           |
|                                 VC5                           |
|                                 VC6                           |
|                                 C6                            |
|                                 C7                            |
|                                 C8                            |
|                                                               |
|    ----------------------------------------------------------  |
|                Total Price     : RM 5513.10                   |
|                Payment Method  : E-Wallet                     |
|                                                               |
|                                                               |
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```

```
                                  ----------------------------------------
                                            Tickets Generated
                                  ----------------------------------------


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|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            | Venue :                  |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                         | National Stadium         |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                         | Bukit Jalil              |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                  | Kuala Lumpur             |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                   |                          |
| GATE  15A              |                                                                                            |                          |
| SEAT  VB4              |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday                |
|                        |                       _____                                  _____                          | December 22, 2023        |
|                        |                    ( ___ )----------------------( ___ )                                  |                          |
| GATE Open 6.30pm       |                     |   |  _   _   ___   ____   |   |                                    |   ------------------   |
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                                    |  | " All MUSIC      |  |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                                    |  |   IS BEAUTIFUL " |  |
| Queue No. 18           |                     |   |   \_/  )_____()_(     |   |                                    |   ------------------   |
|                        |                     |___|                       |___|                                    |                          |
|                        |                    (_____)----------------------(_____)                                  | C  O  O  L  P  L  A  Y   |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            | Venue :                  |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                         | National Stadium         |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                         | Bukit Jalil              |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                  | Kuala Lumpur             |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                   |                          |
| GATE  15A              |                                                                                            |                          |
| SEAT  VB5              |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday                |
```

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|                     |                     _____                     _____                 | December 22, 2023       |
|                     |                ( ___ )-------------------( ___ )                  |                         |
| GATE Open 6.30pm    |                 |   |  _   _   ___   ____   |   |                  |   ------------------    |
|                     |                 |   | \ ( ) / )_ _( )  _)\  |   |                  |  | " All MUSIC      |   |
|                     |                 |   |  )\_/(  _| |_ | '__/ |   |                  |  |   IS BEAUTIFUL " |   |
| Queue No. 17        |                 |   |   \_/  )_____()_(     |   |                  |   ------------------    |
|                     |                 |___|                       |___|                  |                         |
|                     |                (_____)-------------------(_____)                 | C  O  O  L  P  L  A  Y  |
|                     | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                     | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                 |                                                                                                     | Venue :                 |
| RM    788.90        |     \  |  |  |   __| _ _|   __|           _|    |    |                                                  |                         |
|                     |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                                  | National Stadium        |
| **FREE T-SHIRT      |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                                  | Bukit Jalil             |
| **FREE GLOW STICK   |   \__ \  __ / __ | _|     /  _ | \__ \                                                          | Kuala Lumpur            |
|                     |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                            |                         |
| GATE  15A           |                                                                                                     |                         |
| SEAT  VB6           |                          >> COOLPLAY WORLD TOUR <<                                                  | Wednesday               |
|                     |                     _____                     _____                                                   | December 22, 2023       |
|                     |                ( ___ )-------------------( ___ )                                                    |                         |
| GATE Open 6.30pm    |                 |   |  _   _   ___   ____   |   |                                                    |   ------------------    |
|                     |                 |   | \ ( ) / )_ _( )  _)\  |   |                                                    |  | " All MUSIC      |   |
|                     |                 |   |  )\_/(  _| |_ | '__/ |   |                                                    |  |   IS BEAUTIFUL " |   |
| Queue No. 16        |                 |   |   \_/  )_____()_(     |   |                                                    |   ------------------    |
|                     |                 |___|                       |___|                                                    |                         |
|                     |                (_____)-------------------(_____)                                                    | C  O  O  L  P  L  A  Y  |
|                     | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                     | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                 |                                                                                                     | Venue :                 |
| RM    788.90        |     \  |  |  |   __| _ _|   __|           _|    |    |                                                  |                         |
|                     |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                                  | National Stadium        |
| **FREE T-SHIRT      |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                                  | Bukit Jalil             |
| **FREE GLOW STICK   |   \__ \  __ / __ | _|     /  _ | \__ \                                                          | Kuala Lumpur            |
|                     |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                            |                         |
| GATE  15A           |                                                                                                     |                         |
| SEAT  VB7           |                          >> COOLPLAY WORLD TOUR <<                                                  | Wednesday               |
|                     |                     _____                     _____                                                   | December 22, 2023       |
|                     |                ( ___ )-------------------( ___ )                                                    |                         |
| GATE Open 6.30pm    |                 |   |  _   _   ___   ____   |   |                                                    |   ------------------    |
|                     |                 |   | \ ( ) / )_ _( )  _)\  |   |                                                    |  | " All MUSIC      |   |
|                     |                 |   |  )\_/(  _| |_ | '__/ |   |                                                    |  |   IS BEAUTIFUL " |   |
| Queue No. 15        |                 |   |   \_/  )_____()_(     |   |                                                    |   ------------------    |
|                     |                 |___|                       |___|                                                    |                         |
|                     |                (_____)-------------------(_____)                                                    | C  O  O  L  P  L  A  Y  |
|                     | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                     | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                 |                                                                                                     | Venue :                 |
| RM    788.90        |     \  |  |  |   __| _ _|   __|           _|    |    |                                                  |                         |
|                     |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                                  | National Stadium        |
| **FREE T-SHIRT      |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                                  | Bukit Jalil             |
| **FREE GLOW STICK   |   \__ \  __ / __ | _|     /  _ | \__ \                                                          | Kuala Lumpur            |
|                     |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                            |                         |
| GATE  15A           |                                                                                                     |                         |
| SEAT  VC5           |                          >> COOLPLAY WORLD TOUR <<                                                  | Wednesday               |
|                     |                     _____                     _____                                                   | December 22, 2023       |
|                     |                ( ___ )-------------------( ___ )                                                    |                         |
| GATE Open 6.30pm    |                 |   |  _   _   ___   ____   |   |                                                    |   ------------------    |
|                     |                 |   | \ ( ) / )_ _( )  _)\  |   |                                                    |  | " All MUSIC      |   |
|                     |                 |   |  )\_/(  _| |_ | '__/ |   |                                                    |  |   IS BEAUTIFUL " |   |
| Queue No. 14        |                 |   |   \_/  )_____()_(     |   |                                                    |   ------------------    |
|                     |                 |___|                       |___|                                                    |                         |
|                     |                (_____)-------------------(_____)                                                    | C  O  O  L  P  L  A  Y  |
|                     | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
```

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|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP*
| VIP                    |                                                                                  | Venue :
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                          |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                  | National Stadium
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                  | Bukit Jalil
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                          | Kuala Lumpur
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                            |
| GATE  15A              |                                                                                  |
| SEAT  VC6              |                          >> COOLPLAY WORLD TOUR <<                               | Wednesday
|                        |                      _____                      _____                               | December 22, 2023
|                        |                    ( ___ )----------------------( ___ )                           |
| GATE Open 6.30pm       |                     |   |  _    _    ___   ____   |   |                           |    _________________
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                           |   | " All MUSIC      |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                           |   |   IS BEAUTIFUL " |
| Queue No. 13           |                     |   |   \_/  )_____()_(     |   |                           |    -----------------
|                        |                     |___|                       |___|                           |
|                        |                    (_____)----------------------(_____)                          | C  O  O  L  P  L  A  Y
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP*
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| General                |                      **COOLPLAY WORLD TOUR**                                     | Venue :                |
| RM    259.90           |   __  __                  _                  __   _    _                          |                        |
|                        |  |  \/  |                (_)               / _| | |  | |                          | National Stadium       |
|                        |  | \  / | _   _  ___  _   ___    ___  | |_  | | |_ | |__    ___                 | Bukit Jalil            |
|                        |  | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                  | Kuala Lumpur           |
|                        |  | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/                  |                        |
| GATE  15A              |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|                  |                        |
| SEAT  C6               |                                                                                  | Wednesday              |
|                        |              _____          _                                                     | December 22, 2023      |
|                        |             / ____|        | |                                                    |                        |
| GATE Open 7.00pm       |            | (___   _ __  | |__    ___  _ __  ___  ___                          |    _________________   |
|                        |             \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|                          |   | " WHERE MEMORIES |  |
|                        |             ____) || |_) || | | ||  __/| |   |  __/\__ \                          |   |    ARE CREATED " |  |
| Queue No. 12           |            |_____/ | .__/ |_| |_| \___||_|    \___||___/                          |    -----------------   |
|                        |                  | |                                                             |                        |
|                        |                  |_|                                                             | C  O  O  L  P  L  A  Y |
 ------------------------------------------------------------------------------------------------------------------------
```

```
 ------------------------------------------------------------------------------------------------------------------------
| General                |                      **COOLPLAY WORLD TOUR**                                     | Venue :                |
| RM    259.90           |   __  __                  _                  __   _    _                          |                        |
|                        |  |  \/  |                (_)               / _| | |  | |                          | National Stadium       |
|                        |  | \  / | _   _  ___  _   ___    ___  | |_  | | |_ | |__    ___                 | Bukit Jalil            |
|                        |  | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                  | Kuala Lumpur           |
|                        |  | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/                  |                        |
| GATE  15A              |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|                  |                        |
| SEAT  C7               |                                                                                  | Wednesday              |
|                        |              _____          _                                                     | December 22, 2023      |
|                        |             / ____|        | |                                                    |                        |
| GATE Open 7.00pm       |            | (___   _ __  | |__    ___  _ __  ___  ___                          |    _________________   |
|                        |             \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|                          |   | " WHERE MEMORIES |  |
|                        |             ____) || |_) || | | ||  __/| |   |  __/\__ \                          |   |    ARE CREATED " |  |
| Queue No. 11           |            |_____/ | .__/ |_| |_| \___||_|    \___||___/                          |    -----------------   |
|                        |                  | |                                                             |                        |
|                        |                  |_|                                                             | C  O  O  L  P  L  A  Y |
 ------------------------------------------------------------------------------------------------------------------------
```

```
 ------------------------------------------------------------------------------------------------------------------------
| General                |                      **COOLPLAY WORLD TOUR**                                     | Venue :                |
| RM    259.90           |   __  __                  _                  __   _    _                          |                        |
|                        |  |  \/  |                (_)               / _| | |  | |                          | National Stadium       |
|                        |  | \  / | _   _  ___  _   ___    ___  | |_  | | |_ | |__    ___                 | Bukit Jalil            |
|                        |  | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                  | Kuala Lumpur           |
|                        |  | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/                  |                        |
| GATE  15A              |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|                  |                        |
| SEAT  C8               |                                                                                  | Wednesday              |
|                        |              _____          _                                                     | December 22, 2023      |
|                        |             / ____|        | |                                                    |                        |
| GATE Open 7.00pm       |            | (___   _ __  | |__    ___  _ __  ___  ___                          |    _________________   |
|                        |             \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|                          |   | " WHERE MEMORIES |  |
|                        |             ____) || |_) || | | ||  __/| |   |  __/\__ \                          |   |    ARE CREATED " |  |
| Queue No. 10           |            |_____/ | .__/ |_| |_| \___||_|    \___||___/                          |    -----------------   |
|                        |                  | |                                                             |                        |
|                        |                  |_|                                                             | C  O  O  L  P  L  A  Y |
 ------------------------------------------------------------------------------------------------------------------------
```

Sales Report

Order No.	Expected output (after customer order)			
	Category	**Seat left**	**Seat Sold**	**Profit (RM)**
1	VIP	28	2	1577.80
	General	140	0	0.00
	Total Profit: 1577.80 No. of customer: 1 VIP bar (dots): 2 General bar (dots): 0			
2	VIP	26	4	3155.60
	General	135	5	1299.50
	Total Profit: 4455.10 No. of customer: 2 VIP bar (dots): 4 General bar (dots): 5			
3	VIP	24	10	7889.00
	General	137	8	2079.20
	Total Profit: 9968.20 No. of customer: 3 VIP bar (dots): 10 General bar (dots): 8			

```
 ___                  _       ___    _
(  _`\               (_ )    (  _`\(_ )
| ( (_)   _      _    | |    | |_) )| |     _ _   _    _
| |  _  /'_`\  /'_`\  | |    | ,__/'| |  /'_` )( ) ( )
| (_( )( (_) )( (_) ) | |    | |    | | ( (_| || (_) |
(____/''\___/''\___/'(___)   (_)   (___)`\__,_)`\__, |
                                                ( )_| |
                                                `\___/'


    Welcome to concert ticketing system for
   ----------------------------------------
               Cool Play Concert
   ----------------------------------------


              1 -- Seat display
              2 -- Order ticket
              3 -- Sales report
              4 -- Close system

             Select an option: 3
```

Actual Output of Final Sales Report:

```
            / ___|  __ _| | ___  ___  |  _ \ ___ _ __   ___  _ __| |_
            \___ \ / _` | |/ _ \/ __| | |_) / _ \ '_ \ / _ \| '__| __|
             ___) | (_| | |  __/\__ \ |  _ <  __/ |_) | (_) | |  | |_
            |____/ \__,_|_|\___||___/ |_| \_\___| .__/ \___/|_|   \__|
                                                |_|
           Date: 17 December 2023
           Event: CoolPlay Concert
           Sales Person: user1
--------------------------------------------------------------------------------

                      +=======================================+
                      |____  __________  ___  _______  _______|
                      ||  |  |__|__|__|\ ||   [__|__|  |__[__ |
                      ||__|__| \| \|__| \||   ___|  |__|_____]|
                      +=======================================+

   TypeSeat      | PRICE (RM) | TOTAL SEAT | SEAT LEFT | SEAT SOLD | PROFIT (RM)
-----------------|------------|------------|-----------|-----------|------------
      VIP        |     788.90 |         30 |         20|         10|    7889.00
    GENERAL      |     250.90 |        140 |        132|          8|    2079.20
-------------------------------------------------------------------------------
  TOTAL PROFIT(RM)                                                     9968.20
-------------------------------------------------------------------------------
 Number of customer(s): 3

                            *
                            *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                           VIP           General

================================================================================
Note that one * = 1 tickets
```

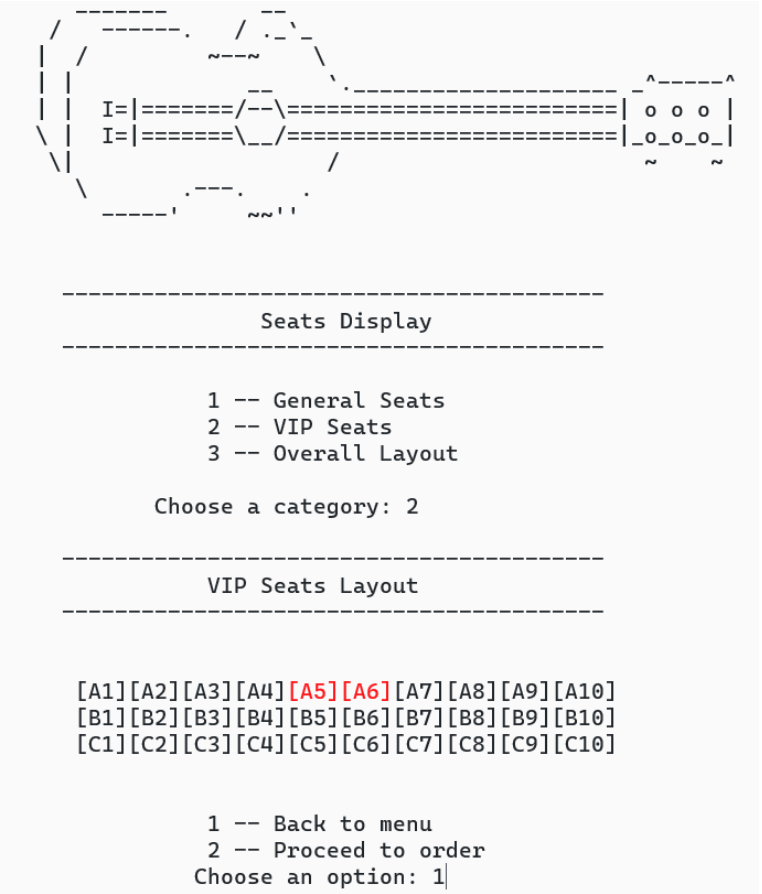
Seats Display

```
 ___                      _       ___   _
(  _`\                   (_ )    (  _`\(_ )
| ( (_)   _       _       | |    | (_) )| |     _ _   _    _
| |  _  /'_`\  /'_`\     | |    | ,__/'| |   /'_` )( ) ( )
| (_( )( (_) )( (_) )   | |    | |    | | ( (_| || (_) |
(____/''\___/''\___/'(___)   (_)   (___)`\__,_)`\__, |
                                                 ( )_| |
                                                 `\___/'


    Welcome to concert ticketing system for
   ----------------------------------------
                Cool Play Concert
   ----------------------------------------


              1 -- Seat display
              2 -- Order ticket
              3 -- Sales report
              4 -- Close system

             Select an option: 1
```

Order No.	Option	Expected Output
1	2 (VIP)	

```
     _______       __
   /   ______.    / ._`_
  |  /          ~---~     \
  | |                 __     `._____________________ _^_____^
  | |  I=|=======/--\=========================| o o o |
  \ |  I=|=======\__/=========================|_o_o_o_|
   \|                     /                      ~    ~
     \        .---.     .
      _____'     ~~''


   ----------------------------------------
                  Seats Display
   ----------------------------------------

              1 -- General Seats
              2 -- VIP Seats
              3 -- Overall Layout

          Choose a category: 2

   ----------------------------------------
              VIP Seats Layout
   ----------------------------------------


    [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
    [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
    [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


              1 -- Back to menu
              2 -- Proceed to order
             Choose an option: 1
```

2	3 (Overall)	``` ---------- Seats Display ---------- 1 -- General Seats 2 -- VIP Seats 3 -- Overall Layout Choose a category: 3 ---------- | | | | | Stage | | | | | ---------- [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10] [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10] [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10] [A1][A2][A3][A4][A5] [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15] [A16][A17][A18][A19][A20] [B1][B2][B3][B4][B5] [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15] [B16][B17][B18][B19][B20] [C1][C2][C3][C4][C5] [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15] [C16][C17][C18][C19][C20] [D1][D2][D3][D4][D5] [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15] [D16][D17][D18][D19][D20] [E1][E2][E3][E4][E5] [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15] [E16][E17][E18][E19][E20] [F1][F2][F3][F4][F5] [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15] [F16][F17][F18][F19][F20] [G1][G2][G3][G4][G5] [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15] [G16][G17][G18][G19][G20] ```
3	3	``` ---------- | | | | | Stage | | | | | ---------- [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10] [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10] [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10] [A1][A2][A3][A4][A5] [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15] [A16][A17][A18][A19][A20] [B1][B2][B3][B4][B5] [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15] [B16][B17][B18][B19][B20] [C1][C2][C3][C4][C5] [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15] [C16][C17][C18][C19][C20] [D1][D2][D3][D4][D5] [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15] [D16][D17][D18][D19][D20] [E1][E2][E3][E4][E5] [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15] [E16][E17][E18][E19][E20] [F1][F2][F3][F4][F5] [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15] [F16][F17][F18][F19][F20] [G1][G2][G3][G4][G5] [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15] [G16][G17][G18][G19][G20] ```

Scenario 2

Order No.	Input				Output	
	Ticket Category		Seat Code	Payment Method	Price Payable	
	VIP	General			Total for each category	Total payment
1	10	0	VIP: [A1] [A2] [A3] [A4] [A5] [A6] [A7] [A8] [A9] [A10]	Credit / Debit card	VIP: 10*788.90 = 7889.00	7889.00
2	2	5	VIP: [A5] [B5] [B6] General: [A10] [A11] [A12] [B11] [B12]	E-Wallet	VIP: 2*788.90 = 1577.80 General: 5*259.90 = 1299.50	1577.80 + 1299.50 = 2877.30
3	3	7	VIP: [A5] [B5] [B6] [C5] [C6] [C7] General: [B14] [B15] [B16] [C15] [C16] [C17] [B17]	Cash	VIP: 3*788.90 = 2366.70 General: 7*259.90 = 1819.30	2366.70 + 1819.30 = 4186.00

Actual Output

Order 1 (Test for input validation):

```
----------------------------------------
            Ticket ordering
----------------------------------------


   No.    Ticket Type          Price
----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: t
Invalid input. Please enter again (1/2): 5
Invalid input. Please enter again (1/2): 1


Please enter amount of ticket to buy: -4
Invalid input, must be positive integer.
Please enter amount of ticket to buy: 10



  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]
```

```
Please enter seat code 1: A1
Please enter seat code 2: A2
Please enter seat code 3: A3
Please enter seat code 4: A4
Please enter seat code 5: A5
Please enter seat code 6: A6
Please enter seat code 7: A7
Please enter seat code 8: A8
Please enter seat code 9: A9
Please enter seat code 10: A10
Do you want add ticket? (Y/N): e
Invalid input. Only Y/N: 0
Invalid input. Only Y/N: N
```

//Prompt error message for category other than ‘1’ and ‘2’
//Prompt error message for negative quantity of tickets
//Prompt error message for input other than ‘Y’/’y’ and ‘N’/’n’

```
 Ticket Category        Price/ticket       Quantity         Amount
*******************************************************************
            VIP              788.90             10        7889.00
-------------------------------------------------------------------
                 Total payment: RM 7889.00
-------------------------------------------------------------------


                 Confirm Payment? (Y/N): l
                Invalid input. Only Y/N: 9
                Invalid input. Only Y/N: Y


     ----------------------------------------
                 Payment Section
     ----------------------------------------


               1 -- Credit/Debit Card
               2 -- Cash
               3 -- E-Wallet

        Choose a payment method: 6
    Invalid input. Please enter again (1/2/3): o
    Invalid input. Please enter again (1/2/3): 1
```

```
Enter credit card number (no space): 12345678901234567
Invalid credit card number. Only 16 digits allowed.
Enter credit card number (no space): 123456789012345
Invalid credit card number. Only 16 digits allowed.
Enter credit card number (no space): 1234567890123456

Expiry Date (MM): 13
Please enter a valid month (1 - 12).
Expiry Date (MM): -2
Please enter a valid month (1 - 12).
Expiry Date (MM): 4
Expiry Date (MM/YY): 4/23
Please enter a valid year (24 - 28).
Expiry Date (MM/YY): 4/25
```

```
Enter name on card: Lee Jun Kit


   Transaction successful.


   Print receipt? (Y/N): Y
```

//Prompt error message for invalid input at confirm payment section
//Prompt error message for invalid payment method input
//Prompt error message for card number when input is not 16 digits.
//Prompt error message when user enters invalid expiry month and year.

```
 ____________________________________________________________
|                                                            |
|          +----------------------------------------+        |
|          |   _____          __  ___  __           |        |
|          |  / ___/__  ___  / / / _ \/ /__ ___ __  |        |
|          | / /__/ _ \/ _ \/ / / ___/ / _ `/ // /  |        |
|          | \___/\___/\___/_/ /_/  /_/\_,_/\_, /   |        |
|          |                                /___/   |        |
|          +----------------------------------------+        |
|                      ** RECEIPT **                         |
|                                                            |
|              Invoice number  : INV0001                     |
|              Date            : 17/12/2023                  |
|   ==========================================================|
|                                                            |
|              Ticket category     Amount    Price (RM)      |
|              VIP                 10        788.90          |
|                                                            |
|              Seat(s) code    :                             |
|                                VA1                         |
|                                VA2                         |
|                                VA3                         |
|                                VA4                         |
|                                VA5                         |
|                                VA6                         |
|                                VA7                         |
|                                VA8                         |
|                                VA9                         |
|                                VA10                        |
|                                                            |
|   ----------------------------------------------------------|
|              Total Price     : RM 7889.00                 |
|              Payment Method  : Credit / Debit Card         |
|              Card Payment Ref: 1234567890123456            |
|                                                            |
 ------------------------------------------------------------


                 Please take the receipt.
```

```
                                  ----------------------------------------
                                            Tickets Generated
                                  ----------------------------------------


 ____________________ | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                |                                                                                                   | Venue :                |
| RM    788.90       |     \  |  |  |   __| _ _|   __|          _|    |    |                                  |                        |
|                    |    |\/ |  |  | \__ \   |   (         _ \   _|     _|    \    -_)                        | National Stadium       |
| **FREE T-SHIRT     |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil            |
| **FREE GLOW STICK  |   \__ \   __ / __ | _|      /  _ | \__ \                                                  | Kuala Lumpur           |
|                    |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                    |                        |
| GATE  15A          |                                                                                                   |                        |
| SEAT  VA1          |                          >> COOLPLAY WORLD TOUR <<                                                | Wednesday              |
|                    |                      _____                                  _____                                | December 22, 2023      |
|                    |                    ( ___ )----------------------------( ___ )                                  |                        |
| GATE Open 6.30pm   |                     |   |  _   _   ___   ____          |   |                                   |  _____________         |
|                    |                     |   | \ ( ) / )_ _( )  _)\         |   |                                   | " All MUSIC      "     |
|                    |                     |   |  )\_/(  _| |_ | '__/         |   |                                   |   IS BEAUTIFUL "       |
| Queue No. 10       |                     |   |   \_/  )_____()_(            |   |                                   |  ------------------    |
|                    |                     |___|                              |___|                                   |                        |
|                    |                    (_____)----------------------------(_____)                                  | C  O  O  L  P  L  A  Y |
|                    | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ----------------------------------------------------------------------------------------------------------------------------


 ____________________ | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                |                                                                                                   | Venue :                |
| RM    788.90       |     \  |  |  |   __| _ _|   __|          _|    |    |                                  |                        |
|                    |    |\/ |  |  | \__ \   |   (         _ \   _|     _|    \    -_)                        | National Stadium       |
| **FREE T-SHIRT     |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil            |
| **FREE GLOW STICK  |   \__ \   __ / __ | _|      /  _ | \__ \                                                  | Kuala Lumpur           |
|                    |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                    |                        |
| GATE  15A          |                                                                                                   |                        |
| SEAT  VA2          |                          >> COOLPLAY WORLD TOUR <<                                                | Wednesday              |
|                    |                      _____                                  _____                                | December 22, 2023      |
|                    |                    ( ___ )----------------------------( ___ )                                  |                        |
| GATE Open 6.30pm   |                     |   |  _   _   ___   ____          |   |                                   |  _____________         |
|                    |                     |   | \ ( ) / )_ _( )  _)\         |   |                                   | " All MUSIC      "     |
|                    |                     |   |  )\_/(  _| |_ | '__/         |   |                                   |   IS BEAUTIFUL "       |
| Queue No. 9        |                     |   |   \_/  )_____()_(            |   |                                   |  ------------------    |
|                    |                     |___|                              |___|                                   |                        |
|                    |                    (_____)----------------------------(_____)                                  | C  O  O  L  P  L  A  Y |
|                    | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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```

```
 _____________________________________________________________________________________________
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                        | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|          _|    |    |                | National Stadium         |
|                      |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)   | Bukit Jalil              |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|     | Kuala Lumpur             |
| **FREE GLOW STICK    |   \__ \   __ / __ | _|     /  _ | \__ \                              |                          |
| GATE  15A            |   ___/ _|  _| _| ___| _|_\ ___| ___/                                |                          |
| SEAT  VA3            |                        >> COOLPLAY WORLD TOUR <<                       | Wednesday                |
|                      |                  ____                                  ____           | December 22, 2023        |
|                      |                ( ___ )--------------------------( ___ )                |                          |
| GATE Open 6.30pm     |                 | |   |  _ _   ___   ____      | |   |                 |  _________________      |
|                      |                 | | \ ( ) / )__ _( )  _)\    | |   |                 | " All MUSIC      "      |
|                      |                 | |  )\_/(  _| |_ | '__/    | |   |                 |   IS BEAUTIFUL "        |
| Queue No. 8          |                 | |   \_/  )_____()_(        | |   |                  ------------------      |
|                      |                 |___|                        |___|                                                |
|                      |               (_____)--------------------------(_____)                 | C  O  O  L  P  L  A  Y   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 -----------------------------------------------------------------------------------------------
```

```
 _____________________________________________________________________________________________
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                        | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|          _|    |    |                | National Stadium         |
|                      |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)   | Bukit Jalil              |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|     | Kuala Lumpur             |
| **FREE GLOW STICK    |   \__ \   __ / __ | _|     /  _ | \__ \                              |                          |
| GATE  15A            |   ___/ _|  _| _| ___| _|_\ ___| ___/                                |                          |
| SEAT  VA4            |                        >> COOLPLAY WORLD TOUR <<                       | Wednesday                |
|                      |                  ____                                  ____           | December 22, 2023        |
|                      |                ( ___ )--------------------------( ___ )                |                          |
| GATE Open 6.30pm     |                 | |   |  _ _   ___   ____      | |   |                 |  _________________      |
|                      |                 | | \ ( ) / )__ _( )  _)\    | |   |                 | " All MUSIC      "      |
|                      |                 | |  )\_/(  _| |_ | '__/    | |   |                 |   IS BEAUTIFUL "        |
| Queue No. 7          |                 | |   \_/  )_____()_(        | |   |                  ------------------      |
|                      |                 |___|                        |___|                                                |
|                      |               (_____)--------------------------(_____)                 | C  O  O  L  P  L  A  Y   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                        | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|          _|    |    |                | National Stadium         |
|                      |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)   | Bukit Jalil              |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|     | Kuala Lumpur             |
| **FREE GLOW STICK    |   \__ \   __ / __ | _|     /  _ | \__ \                              |                          |
| GATE  15A            |   ___/ _|  _| _| ___| _|_\ ___| ___/                                |                          |
| SEAT  VA5            |                        >> COOLPLAY WORLD TOUR <<                       | Wednesday                |
|                      |                  ____                                  ____           | December 22, 2023        |
|                      |                ( ___ )--------------------------( ___ )                |                          |
| GATE Open 6.30pm     |                 | |   |  _ _   ___   ____      | |   |                 |  _________________      |
|                      |                 | | \ ( ) / )__ _( )  _)\    | |   |                 | " All MUSIC      "      |
|                      |                 | |  )\_/(  _| |_ | '__/    | |   |                 |   IS BEAUTIFUL "        |
| Queue No. 6          |                 | |   \_/  )_____()_(        | |   |                  ------------------      |
|                      |                 |___|                        |___|                                                |
|                      |               (_____)--------------------------(_____)                 | C  O  O  L  P  L  A  Y   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                        | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|          _|    |    |                | National Stadium         |
|                      |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)   | Bukit Jalil              |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|     | Kuala Lumpur             |
| **FREE GLOW STICK    |   \__ \   __ / __ | _|     /  _ | \__ \                              |                          |
| GATE  15A            |   ___/ _|  _| _| ___| _|_\ ___| ___/                                |                          |
| SEAT  VA6            |                        >> COOLPLAY WORLD TOUR <<                       | Wednesday                |
|                      |                  ____                                  ____           | December 22, 2023        |
|                      |                ( ___ )--------------------------( ___ )                |                          |
| GATE Open 6.30pm     |                 | |   |  _ _   ___   ____      | |   |                 |  _________________      |
|                      |                 | | \ ( ) / )__ _( )  _)\    | |   |                 | " All MUSIC      "      |
|                      |                 | |  )\_/(  _| |_ | '__/    | |   |                 |   IS BEAUTIFUL "        |
| Queue No. 5          |                 | |   \_/  )_____()_(        | |   |                  ------------------      |
|                      |                 |___|                        |___|                                                |
|                      |               (_____)--------------------------(_____)                 | C  O  O  L  P  L  A  Y   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                        | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|          _|    |    |                | National Stadium         |
|                      |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)   | Bukit Jalil              |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|     | Kuala Lumpur             |
| **FREE GLOW STICK    |   \__ \   __ / __ | _|     /  _ | \__ \                              |                          |
| GATE  15A            |   ___/ _|  _| _| ___| _|_\ ___| ___/                                |                          |
| SEAT  VA7            |                        >> COOLPLAY WORLD TOUR <<                       | Wednesday                |
|                      |                  ____                                  ____           | December 22, 2023        |
|                      |                ( ___ )--------------------------( ___ )                |                          |
| GATE Open 6.30pm     |                 | |   |  _ _   ___   ____      | |   |                 |  _________________      |
|                      |                 | | \ ( ) / )__ _( )  _)\    | |   |                 | " All MUSIC      "      |
|                      |                 | |  )\_/(  _| |_ | '__/    | |   |                 |   IS BEAUTIFUL "        |
| Queue No. 4          |                 | |   \_/  )_____()_(        | |   |                  ------------------      |
|                      |                 |___|                        |___|                                                |
|                      |               (_____)--------------------------(_____)                 | C  O  O  L  P  L  A  Y   |
```

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|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                   |                                                                                            | Venue :                  |
| RM    788.90          |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                          |
|                       |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)                        | National Stadium         |
| **FREE T-SHIRT        |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil              |
| **FREE GLOW STICK     |   \__ \   __ / __ | _|      /   _ | \__ \                                                | Kuala Lumpur             |
|                       |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                  |                          |
| GATE  15A             |                                                                                            |                          |
| SEAT  VA8             |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday                |
|                       |                      _____                                  _____                          | December 22, 2023        |
|                       |                    ( ___ )----------------------( ___ )                                  |                          |
| GATE Open 6.30pm      |                     |   |  _   _   ___   ____   |   |                                   |  _________________       |
|                       |                     |   | \ ( ) / )_ _( )  _)\  |   |                                   | " All MUSIC      "       |
|                       |                     |   |  )\_/(  _| |_ | '__/  |   |                                   |   IS BEAUTIFUL "         |
| Queue No. 3           |                     |   |   \_/  )_____()_(     |   |                                   |  ------------------      |
|                       |                     |___|                       |___|                                   |                          |
|                       |                    (_____)----------------------(_____)                                  | C  O  O  L  P  L  A  Y   |
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                   |                                                                                            | Venue :                  |
| RM    788.90          |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                          |
|                       |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)                        | National Stadium         |
| **FREE T-SHIRT        |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil              |
| **FREE GLOW STICK     |   \__ \   __ / __ | _|      /   _ | \__ \                                                | Kuala Lumpur             |
|                       |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                  |                          |
| GATE  15A             |                                                                                            |                          |
| SEAT  VA9             |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday                |
|                       |                      _____                                  _____                          | December 22, 2023        |
|                       |                    ( ___ )----------------------( ___ )                                  |                          |
| GATE Open 6.30pm      |                     |   |  _   _   ___   ____   |   |                                   |  _________________       |
|                       |                     |   | \ ( ) / )_ _( )  _)\  |   |                                   | " All MUSIC      "       |
|                       |                     |   |  )\_/(  _| |_ | '__/  |   |                                   |   IS BEAUTIFUL "         |
| Queue No. 2           |                     |   |   \_/  )_____()_(     |   |                                   |  ------------------      |
|                       |                     |___|                       |___|                                   |                          |
|                       |                    (_____)----------------------(_____)                                  | C  O  O  L  P  L  A  Y   |
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                   |                                                                                            | Venue :                  |
| RM    788.90          |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                          |
|                       |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)                        | National Stadium         |
| **FREE T-SHIRT        |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil              |
| **FREE GLOW STICK     |   \__ \   __ / __ | _|      /   _ | \__ \                                                | Kuala Lumpur             |
|                       |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                  |                          |
| GATE  15A             |                                                                                            |                          |
| SEAT  VA10            |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday                |
|                       |                      _____                                  _____                          | December 22, 2023        |
|                       |                    ( ___ )----------------------( ___ )                                  |                          |
| GATE Open 6.30pm      |                     |   |  _   _   ___   ____   |   |                                   |  _________________       |
|                       |                     |   | \ ( ) / )_ _( )  _)\  |   |                                   | " All MUSIC      "       |
|                       |                     |   |  )\_/(  _| |_ | '__/  |   |                                   |   IS BEAUTIFUL "         |
| Queue No. 1           |                     |   |   \_/  )_____()_(     |   |                                   |  ------------------      |
|                       |                     |___|                       |___|                                   |                          |
|                       |                    (_____)----------------------(_____)                                  | C  O  O  L  P  L  A  Y   |
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 -----------------------------------------------------------------------------------------------------------------------
```

```
Press [Y] if done ... ...o
    Enter [Y] only: 5
    Enter [Y] only: y
```

//Prompt error message for input other than ‘Y’/’y’

Order 2 (Test for buying seats reserved - A5) :

```
 ----------------------------------------
            Ticket ordering
 ----------------------------------------


   No.    Ticket Type          Price
 ----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 2



  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


        Please enter seat code 1: A5
        Seat not available, choose another: B5
        Please enter seat code 2: B6
        Do you want add ticket? (Y/N): y
```

//Prompt error message saying seats not available and ask for valid seat code again.

```
                          ----------------------------------------
                                     Ticket ordering
                          ----------------------------------------


                             No.    Ticket Type          Price
                          ----------------------------------------
                               1            VIP         788.90
                               2        General         259.90


                         Please enter the category of ticket: 2


                         Please enter amount of ticket to buy: 5



     [A1][A2][A3][A4][A5]   [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15]   [A16][A17][A18][A19][A20]
     [B1][B2][B3][B4][B5]   [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15]   [B16][B17][B18][B19][B20]
     [C1][C2][C3][C4][C5]   [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15]   [C16][C17][C18][C19][C20]
     [D1][D2][D3][D4][D5]   [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15]   [D16][D17][D18][D19][D20]
     [E1][E2][E3][E4][E5]   [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15]   [E16][E17][E18][E19][E20]
     [F1][F2][F3][F4][F5]   [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15]   [F16][F17][F18][F19][F20]
     [G1][G2][G3][G4][G5]   [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15]   [G16][G17][G18][G19][G20]


                                 Please enter seat code 1: A10
                                 Please enter seat code 2: A11
                                 Please enter seat code 3: A12
                                 Please enter seat code 4: B11
                                 Please enter seat code 5: B12
                                 Do you want add ticket? (Y/N): n
```

```
 Ticket Category        Price/ticket       Quantity         Amount
*********************************************************************
            VIP              788.90             2        1577.80
        General              259.90             5        1299.50
---------------------------------------------------------------------
                 Total payment: RM 2877.30
---------------------------------------------------------------------


                 Confirm Payment? (Y/N): y


     ----------------------------------------
                  Payment Section
     ----------------------------------------


               1 -- Credit/Debit Card
               2 -- Cash
               3 -- E-Wallet

       Choose a payment method: 3


         Transaction successful.


          Print receipt? (Y/N): n
```

```
                                        ----------------------------------------
                                                 Tickets Generated
                                        ----------------------------------------


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|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                                            |
| RM    788.90         |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |
|                      |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)                         |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                         |
| **FREE GLOW STICK    |   \__ \   __ / __ | _|     /  _ | \__ \                                                |
|                      |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                  |
| GATE  15A            |                                                                                            |
| SEAT  VB5            |                        >> COOLPLAY WORLD TOUR <<                                           |
|                      |                     _____                                  _____                            |
|                      |                   ( ___ )----------------------( ___ )                                   |
| GATE Open 6.30pm     |                     |   |   _   _   ___   ____   |   |                                     |
|                      |                     |   | \ ( ) / )_ _( )  _)\   |   |                                     |
|                      |                     |   |  )\_/(  _| |_ | '__/   |   |                                     |
| Queue No. 17         |                     |   |   \_/  )_____()_(      |   |                                     |
|                      |                     |___|                        |___|                                     |
|                      |                   (_____)----------------------(_____)                                   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                                            |
| RM    788.90         |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |
|                      |    |\/ |  |  | \__ \   |   (          _ \   _|     _|    \    -_)                         |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                         |
| **FREE GLOW STICK    |   \__ \   __ / __ | _|     /  _ | \__ \                                                |
|                      |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                  |
| GATE  15A            |                                                                                            |
| SEAT  VB6            |                        >> COOLPLAY WORLD TOUR <<                                           |
|                      |                     _____                                  _____                            |
|                      |                   ( ___ )----------------------( ___ )                                   |
| GATE Open 6.30pm     |                     |   |   _   _   ___   ____   |   |                                     |
|                      |                     |   | \ ( ) / )_ _( )  _)\   |   |                                     |
|                      |                     |   |  )\_/(  _| |_ | '__/   |   |                                     |
| Queue No. 16         |                     |   |   \_/  )_____()_(      |   |                                     |
|                      |                     |___|                        |___|                                     |
|                      |                   (_____)----------------------(_____)                                   |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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| General              |                        **COOLPLAY WORLD TOUR**                                             |
| RM    259.90         |                                                                                            |
|                      |   __ __                 _                  __   _   _                                      |
|                      |  |  \/  |              (_)               / _| | |  | |                                      |
|                      |  | \  / |  _   _   ___   _   ___   ___  | |_  | |_ | |__    ___                          |
|                      |  | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                          |
|                      |  | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | || __/                          |
| GATE  15A            |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|                          |
| SEAT  A10            |                                                                                            |
|                      |                 _____          _                                                           |
```

```
| GATE Open 7.00pm |
| Queue No. 15     |
```

```
| General          |                    **COOLPLAY WORLD TOUR**                    | Venue :
| RM    259.90     |
| GATE  15A        |                                                               | National Stadium
| SEAT  A11        |                                                               | Bukit Jalil
| GATE Open 7.00pm |                                                               | Kuala Lumpur
| Queue No. 14     |                                                               | Wednesday
|                  |                                                               | December 22, 2023
```

```
| General          |                    **COOLPLAY WORLD TOUR**                    | Venue :
| RM    259.90     |
| GATE  15A        |                                                               | National Stadium
| SEAT  A12        |                                                               | Bukit Jalil
| GATE Open 7.00pm |                                                               | Kuala Lumpur
| Queue No. 13     |                                                               | Wednesday
|                  |                                                               | December 22, 2023
```

```
| General          |                    **COOLPLAY WORLD TOUR**                    | Venue :
| RM    259.90     |
| GATE  15A        |                                                               | National Stadium
| SEAT  B11        |                                                               | Bukit Jalil
| GATE Open 7.00pm |                                                               | Kuala Lumpur
| Queue No. 12     |                                                               | Wednesday
|                  |                                                               | December 22, 2023
```

```
| General          |                    **COOLPLAY WORLD TOUR**                    | Venue :
| RM    259.90     |
| GATE  15A        |                                                               | National Stadium
| SEAT  B12        |                                                               | Bukit Jalil
| GATE Open 7.00pm |                                                               | Kuala Lumpur
| Queue No. 11     |                                                               | Wednesday
|                  |                                                               | December 22, 2023
```

```
| " WHERE MEMORIES |
|    ARE CREATED " |
 ------------------
| C  O  O  L  P  L  A  Y |
```

```
Press [Y] if done ... ...
```

Order 3 (Test for buying seats reserved - A5, B5, B6) :

```
 ----------------------------------------
            Ticket ordering
 ----------------------------------------


   No.    Ticket Type          Price
 ----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 3



  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


        Please enter seat code 1: A5
        Seat not available, choose another: B5
        Seat not available, choose another: B6
        Seat not available, choose another: C5
        Please enter seat code 2: C6
        Please enter seat code 3: C7
        Do you want add ticket? (Y/N): y
```

```
                         ----------------------------------------
                                    Ticket ordering
                         ----------------------------------------


                            No.    Ticket Type          Price
                         ----------------------------------------
                              1            VIP         788.90
                              2        General         259.90


                        Please enter the category of ticket: 2


                        Please enter amount of ticket to buy: 7



   [A1][A2][A3][A4][A5]   [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15]   [A16][A17][A18][A19][A20]
   [B1][B2][B3][B4][B5]   [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15]   [B16][B17][B18][B19][B20]
   [C1][C2][C3][C4][C5]   [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15]   [C16][C17][C18][C19][C20]
   [D1][D2][D3][D4][D5]   [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15]   [D16][D17][D18][D19][D20]
   [E1][E2][E3][E4][E5]   [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15]   [E16][E17][E18][E19][E20]
   [F1][F2][F3][F4][F5]   [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15]   [F16][F17][F18][F19][F20]
   [G1][G2][G3][G4][G5]   [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15]   [G16][G17][G18][G19][G20]


                                Please enter seat code 1: B14
                                Please enter seat code 2: b15
                                Please enter seat code 3: b16
                                Please enter seat code 4: c15
                                Please enter seat code 5: c16
                                Please enter seat code 6: c17
                                Please enter seat code 7: b17
                                Do you want add ticket? (Y/N): n
```

```
  Ticket Category        Price/ticket       Quantity         Amount
*********************************************************************
             VIP              788.90              3        2366.70
         General              259.90              7        1819.30
---------------------------------------------------------------------
                 Total payment: RM 4186.00
---------------------------------------------------------------------


                 Confirm Payment? (Y/N): Y


     ----------------------------------------
                  Payment Section
     ----------------------------------------


               1 -- Credit/Debit Card
               2 -- Cash
               3 -- E-Wallet

        Choose a payment method: 2
           Enter cash paid (RM): -5
      Please enter a positive value.
           Enter cash paid (RM): 4000
           Payment not enough.
                Pay more RM186.00
           Enter cash paid (RM): 4200
               Balance: RM 14.00


           Payment succeed.


           Print receipt? (Y/N): Y
```

//Negative value and not sufficient payment will prompt error message
//If cash paid more than actual amount, display balance (4200.00 - 4186.00 = 14.00)

```
 ---------------------------------------------------------------
|                                                               |
|          +-----------------------------------------+          |
|          |    _____          __  ___  __           |          |
|          |   / ___/__  ___  / / / _ \/ /__ ___ __  |          |
|          |  / /__/ _ \/ _ \/ / / ___/ / _ `/ // /  |          |
|          |  \___/\___/\___/_/ /_/  /_/\_,_/\_, /   |          |
|          |                                /___/    |          |
|          +-----------------------------------------+          |
|                       ** RECEIPT **                           |
|                                                               |
|              Invoice number  : INV0003                        |
|              Date            : 17/12/2023                     |
|   ========================================================    |
|                                                               |
|   Ticket category     Qty       Price (RM)     Amount         |
|   VIP                 3         788.90         2366.70        |
|   General             7         259.90         1819.30        |
|                                                               |
|              Seat(s) code    :                                |
|                               VC5                             |
|                               VC6                             |
|                               VC7                             |
|                               B14                             |
|                               B15                             |
|                               B16                             |
|                               B17                             |
|                               C15                             |
|                               C16                             |
|                               C17                             |
|                                                               |
|   ---------------------------------------------------------   |
|              Total Price     : RM 4186.00                     |
|              Payment Method  : Cash                           |
|              Balance : RM 14.0                                |
|                                                               |
 ---------------------------------------------------------------
```

```
                                        ----------------------------------------
                                                  Tickets Generated
                                        ----------------------------------------


 ____________________________________________________________________________________________________________________
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |
|                        |    |\/ |  |  | \__ \   |   (         _ \   _|     _|    \    -_)                          |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                          |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                  |
|                        |   ___/ _|  _| _| ___| _|_\ ___| ___/                                                      |
| GATE  15A              |                                                                                            |
| SEAT  VC5              |                        >> COOLPLAY WORLD TOUR <<                                           |
|                        |                    _____                                  _____                              |
|                        |                  ( ___ )----------------------------( ___ )                                |
| GATE Open 6.30pm       |                   |   |   _   _   ___   ____          |   |                                |
|                        |                   |   | \ ( ) / )_ _( )  _)\          |   |                                |
|                        |                   |   |  )\_/(  _| |_ | '__/          |   |                                |
| Queue No. 27           |                   |   |   \_/  )_____()_(             |   |                                |
|                        |                   |___|                               |___|                                |
|                        |                  (_____)----------------------------(_____)                                |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------------------


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|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |
|                        |    |\/ |  |  | \__ \   |   (         _ \   _|     _|    \    -_)                          |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                          |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                  |
|                        |   ___/ _|  _| _| ___| _|_\ ___| ___/                                                      |
| GATE  15A              |                                                                                            |
| SEAT  VC6              |                        >> COOLPLAY WORLD TOUR <<                                           |
|                        |                    _____                                  _____                              |
|                        |                  ( ___ )----------------------------( ___ )                                |
| GATE Open 6.30pm       |                   |   |   _   _   ___   ____          |   |                                |
|                        |                   |   | \ ( ) / )_ _( )  _)\          |   |                                |
|                        |                   |   |  )\_/(  _| |_ | '__/          |   |                                |
| Queue No. 26           |                   |   |   \_/  )_____()_(             |   |                                |
|                        |                   |___|                               |___|                                |
|                        |                  (_____)----------------------------(_____)                                |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |
|                        |    |\/ |  |  | \__ \   |   (         _ \   _|     _|    \    -_)                          |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                          |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                  |
|                        |   ___/ _|  _| _| ___| _|_\ ___| ___/                                                      |
| GATE  15A              |                                                                                            |
```

```
| SEAT  VC7              |                      >> COOLPLAY WORLD TOUR <<                  | Wednesday                |
|                        |                  _____                   _____                  | December 22, 2023        |
|                        |                ( ___ )-------------------( ___ )                |                          |
| GATE Open 6.30pm       |                 | |   _   _   ___   ____   | |                 |  _________________      |
|                        |                 | | \ ( ) / )_ _( )  _)\   | |                 | " All MUSIC       "  |
|                        |                 | |  )\_/(  _| |_ | '__/   | |                 |   IS BEAUTIFUL "  |
| Queue No. 25           |                 | |   \_/  )_____()_(      | |                 |  ------------------     |
|                        |                 |___|                     |___|                 |                          |
|                        |                (_____)-------------------(_____)                | C  O  O  L  P  L  A  Y   |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
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```

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 _____________________________________________________________________________________________________________________
| General                |                    **COOLPLAY WORLD TOUR**                      | Venue :                  |
| RM    259.90           |                                                                 |                          |
|                        |  __ __                  _                 __   _   _            |                          |
|                        |  \/  |                (_)               / _| | |  | |           | National Stadium         |
|                        |  \  / |  _   _  ___   _   ___    ___   | |_  | |_ | |__   ___  | Bukit Jalil              |
|                        |  |\/| | | | |/ __|| | / __|  / _ \  |  _| | __|| '_ \  / _ \ | Kuala Lumpur             |
|                        |  |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/ |                          |
| GATE  15A              |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___| |                          |
| SEAT  B14              |                                                                 | Wednesday                |
|                        |                 ____          _                                 | December 22, 2023        |
|                        |                / ___|        | |                                |                          |
| GATE Open 7.00pm       |               | (___   _ __  | |__    ___  _ __   ___  ___    |  _________________      |
|                        |                \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|   | " WHERE MEMORIES "  |
|                        |                ___) || |_) || | | ||  __/| |   |  __/\__ \   |    ARE CREATED "  |
| Queue No. 24           |               |____/ | .__/ |_| |_| \___||_|    \___||___/   |  ------------------     |
|                        |                      | |                                     |                          |
|                        |                      |_|                                     | C  O  O  L  P  L  A  Y   |
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```

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| General                |                    **COOLPLAY WORLD TOUR**                      | Venue :                  |
| RM    259.90           |                                                                 |                          |
|                        |  __ __                  _                 __   _   _            |                          |
|                        |  \/  |                (_)               / _| | |  | |           | National Stadium         |
|                        |  \  / |  _   _  ___   _   ___    ___   | |_  | |_ | |__   ___  | Bukit Jalil              |
|                        |  |\/| | | | |/ __|| | / __|  / _ \  |  _| | __|| '_ \  / _ \ | Kuala Lumpur             |
|                        |  |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/ |                          |
| GATE  15A              |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___| |                          |
| SEAT  B15              |                                                                 | Wednesday                |
|                        |                 ____          _                                 | December 22, 2023        |
|                        |                / ___|        | |                                |                          |
| GATE Open 7.00pm       |               | (___   _ __  | |__    ___  _ __   ___  ___    |  _________________      |
|                        |                \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|   | " WHERE MEMORIES "  |
|                        |                ___) || |_) || | | ||  __/| |   |  __/\__ \   |    ARE CREATED "  |
| Queue No. 23           |               |____/ | .__/ |_| |_| \___||_|    \___||___/   |  ------------------     |
|                        |                      | |                                     |                          |
|                        |                      |_|                                     | C  O  O  L  P  L  A  Y   |
 ---------------------------------------------------------------------------------------------------------------------------
```

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 _____________________________________________________________________________________________________________________
| General                |                    **COOLPLAY WORLD TOUR**                      | Venue :                  |
| RM    259.90           |                                                                 |                          |
|                        |  __ __                  _                 __   _   _            |                          |
|                        |  \/  |                (_)               / _| | |  | |           | National Stadium         |
|                        |  \  / |  _   _  ___   _   ___    ___   | |_  | |_ | |__   ___  | Bukit Jalil              |
|                        |  |\/| | | | |/ __|| | / __|  / _ \  |  _| | __|| '_ \  / _ \ | Kuala Lumpur             |
|                        |  |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/ |                          |
| GATE  15A              |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___| |                          |
| SEAT  B16              |                                                                 | Wednesday                |
|                        |                 ____          _                                 | December 22, 2023        |
```

```
|                        |                / ___|        | |                                |                          |
| GATE Open 7.00pm       |               | (___   _ __  | |__    ___  _ __   ___  ___    |  _________________      |
|                        |                \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|   | " WHERE MEMORIES "  |
|                        |                ___) || |_) || | | ||  __/| |   |  __/\__ \   |    ARE CREATED "  |
| Queue No. 22           |               |____/ | .__/ |_| |_| \___||_|    \___||___/   |  ------------------     |
|                        |                      | |                                     |                          |
|                        |                      |_|                                     | C  O  O  L  P  L  A  Y   |
 ---------------------------------------------------------------------------------------------------------------------------
```

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 _____________________________________________________________________________________________________________________
| General                |                    **COOLPLAY WORLD TOUR**                      | Venue :                  |
| RM    259.90           |                                                                 |                          |
|                        |  __ __                  _                 __   _   _            |                          |
|                        |  \/  |                (_)               / _| | |  | |           | National Stadium         |
|                        |  \  / |  _   _  ___   _   ___    ___   | |_  | |_ | |__   ___  | Bukit Jalil              |
|                        |  |\/| | | | |/ __|| | / __|  / _ \  |  _| | __|| '_ \  / _ \ | Kuala Lumpur             |
|                        |  |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/ |                          |
| GATE  15A              |  |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___| |                          |
| SEAT  B17              |                                                                 | Wednesday                |
|                        |                 ____          _                                 | December 22, 2023        |
|                        |                / ___|        | |                                |                          |
| GATE Open 7.00pm       |               | (___   _ __  | |__    ___  _ __   ___  ___    |  _________________      |
|                        |                \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|   | " WHERE MEMORIES "  |
|                        |                ___) || |_) || | | ||  __/| |   |  __/\__ \   |    ARE CREATED "  |
| Queue No. 21           |               |____/ | .__/ |_| |_| \___||_|    \___||___/   |  ------------------     |
|                        |                      | |                                     |                          |
|                        |                      |_|                                     | C  O  O  L  P  L  A  Y   |
 ---------------------------------------------------------------------------------------------------------------------------
```

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 ____________________________________________________________________________________________________
| General                |                    **COOLPLAY WORLD TOUR**                    | Venue :                |
| RM    259.90           |                                                               |                        |
|                        |  __  __            _              __   _   _                 |                        |
|                        | |  \/  |          (_)            / _| | |  | |                | National Stadium       |
|                        | | \  / |  _   _  ___  _   ___   ___  | |_  | |_| | |__   ___  | Bukit Jalil            |
|                        | | |\/| | | | | |/ __|| | / __|  / _ \ |  _| | __| '_ \  / _ \ | Kuala Lumpur           |
|                        | | |  | | |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/ |                        |
| GATE  15A              | |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___| |                        |
| SEAT  C15              |                                                               | Wednesday              |
|                        |            _____        _                                    | December 22, 2023      |
|                        |           / ____|      | |                                   |                        |
| GATE Open 7.00pm       |          | (___   _ __  | |__   ___  _ __  ___  ___          |   _________________    |
|                        |           \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|         |  | " WHERE MEMORIES "  |
|                        |           ____) || |_) || | | ||  __/| |  |  __/\__ \         |  |    ARE CREATED "    |
| Queue No. 20           |          |_____/ | .__/ |_| |_| \___||_|   \___||___/         |   -----------------    |
|                        |                  | |                                         |                        |
|                        |                  |_|                                         | C  O  O  L  P  L  A  Y |
 ----------------------------------------------------------------------------------------------------

 ____________________________________________________________________________________________________
| General                |                    **COOLPLAY WORLD TOUR**                    | Venue :                |
| RM    259.90           |                                                               |                        |
|                        |  __  __            _              __   _   _                 |                        |
|                        | |  \/  |          (_)            / _| | |  | |                | National Stadium       |
|                        | | \  / |  _   _  ___  _   ___   ___  | |_  | |_| | |__   ___  | Bukit Jalil            |
|                        | | |\/| | | | | |/ __|| | / __|  / _ \ |  _| | __| '_ \  / _ \ | Kuala Lumpur           |
|                        | | |  | | |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/ |                        |
| GATE  15A              | |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___| |                        |
| SEAT  C16              |                                                               | Wednesday              |
|                        |            _____        _                                    | December 22, 2023      |
|                        |           / ____|      | |                                   |                        |
| GATE Open 7.00pm       |          | (___   _ __  | |__   ___  _ __  ___  ___          |   _________________    |
|                        |           \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|         |  | " WHERE MEMORIES "  |
|                        |           ____) || |_) || | | ||  __/| |  |  __/\__ \         |  |    ARE CREATED "    |
| Queue No. 19           |          |_____/ | .__/ |_| |_| \___||_|   \___||___/         |   -----------------    |
|                        |                  | |                                         |                        |
|                        |                  |_|                                         | C  O  O  L  P  L  A  Y |
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 ____________________________________________________________________________________________________
| General                |                    **COOLPLAY WORLD TOUR**                    | Venue :                |
| RM    259.90           |                                                               |                        |
|                        |  __  __            _              __   _   _                 |                        |
|                        | |  \/  |          (_)            / _| | |  | |                | National Stadium       |
|                        | | \  / |  _   _  ___  _   ___   ___  | |_  | |_| | |__   ___  | Bukit Jalil            |
|                        | | |\/| | | | | |/ __|| | / __|  / _ \ |  _| | __| '_ \  / _ \ | Kuala Lumpur           |
|                        | | |  | | |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/ |                        |
| GATE  15A              | |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___| |                        |
| SEAT  C17              |                                                               | Wednesday              |
|                        |            _____        _                                    | December 22, 2023      |
|                        |           / ____|      | |                                   |                        |
| GATE Open 7.00pm       |          | (___   _ __  | |__   ___  _ __  ___  ___          |   _________________    |
|                        |           \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|         |  | " WHERE MEMORIES "  |
|                        |           ____) || |_) || | | ||  __/| |  |  __/\__ \         |  |    ARE CREATED "    |
| Queue No. 18           |          |_____/ | .__/ |_| |_| \___||_|   \___||___/         |   -----------------    |
|                        |                  | |                                         |                        |
|                        |                  |_|                                         | C  O  O  L  P  L  A  Y |
 ----------------------------------------------------------------------------------------------------

                                Press [Y] if done ... ...
```
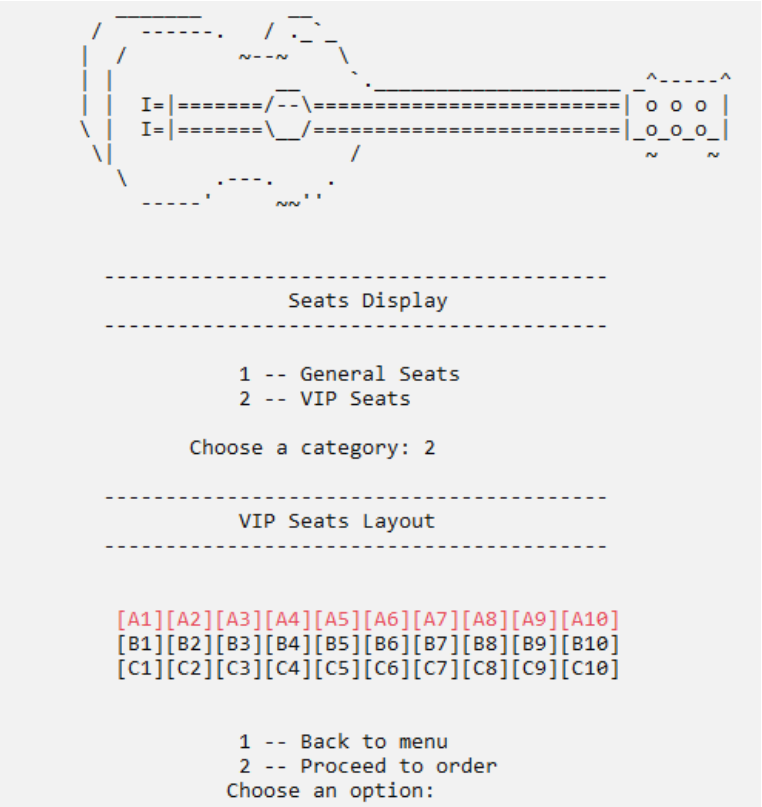
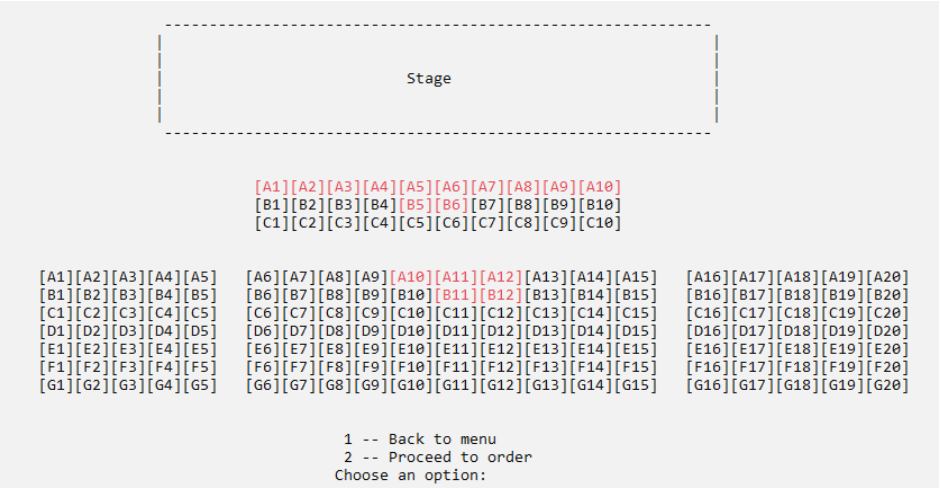

Sales Report

Order No.	Expected output (after customer order)			
	Category	**Seat left**	**Seat Sold**	**Profit (RM)**
1	VIP	20	10	7889.00
	General	140	0	0.00
	Total Profit: 7889.00 No. of customer: 1 VIP bar (dots): 10 General bar (dots): 0			
2	VIP	18	12	9466.80
	General	135	5	1299.50
	Total Profit: 10766.30 No. of customer: 2 VIP bar (dots): 12 General bar (dots): 5			
3	VIP	15	15	11833.50
	General	128	12	3118.80
	Total Profit: 14952.30 No. of customer: 3 VIP bar (dots): 15 General bar (dots): 12			

Actual Output of Final Sales Report:

```
==============================================================================

            ____       _           ____                      _
           / ___|  __ _| |___  ___ |  _ \ ___ _ __  ___  _ __| |_
           \___ \ / _` | |/ _ \/ __| | |_) / _ \ '_ \ / _ \| '__| __|
            ___) | (_| | |  __/\__ \ |  _ <  __/ |_) | (_) | |  | |_
           |____/ \__,_|_|\___||___/ |_| \_\___| .__/ \___/|_|   \__|
                                                |_|
          Date: 17 December 2023
          Event: CoolPlay Concert
          Sales Person: user1
------------------------------------------------------------------------------

                      +=======================================+
                      |____  _________  ___  _______  _______|
                      ||  |  |__|__|__|\ ||   [__|__|  |__[__ |
                      ||__|__| \| \|__| \||   ___|  |__|_____]|
                      +=======================================+

   TypeSeat      | PRICE (RM) | TOTAL SEAT | SEAT LEFT | SEAT SOLD | PROFIT (RM)
-----------------|------------|------------|-----------|-----------|------------
      VIP        |     788.90 |         30 |        15|         15|   11833.50
    GENERAL      |     250.90 |        140 |       128|         12|    3118.80
-----------------------------------------------------------------------------
  TOTAL PROFIT(RM)                                                    14952.30
-----------------------------------------------------------------------------
 Number of customer(s): 3

                            *
                            *
                            *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                            *                *
                           VIP            General

==============================================================================
Note that one * = 1 tickets


                      Press [Y] back to menu :
```

Seats Display

Order No.	Option	Expected Output
1	2 (VIP)	
2	3 (Overall)	

3	3	``` ---------- | | | | | Stage | | | | | ---------- [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10] [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10] [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10] [A1][A2][A3][A4][A5] [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15] [A16][A17][A18][A19][A20] [B1][B2][B3][B4][B5] [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15] [B16][B17][B18][B19][B20] [C1][C2][C3][C4][C5] [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15] [C16][C17][C18][C19][C20] [D1][D2][D3][D4][D5] [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15] [D16][D17][D18][D19][D20] [E1][E2][E3][E4][E5] [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15] [E16][E17][E18][E19][E20] [F1][F2][F3][F4][F5] [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15] [F16][F17][F18][F19][F20] [G1][G2][G3][G4][G5] [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15] [G16][G17][G18][G19][G20] 1 -- Back to menu 2 -- Proceed to order Choose an option: ```

Scenario 3

Order No.	Input				Output	
	Ticket Category		Seat Code	Payment Method	Price Payable	
	VIP	General			Total for each category	Total payment
1	17	0	VIP: [A1] [A2] [A3] [A4] [A5] [A6] [A7] [A8] [A9] [A10] [A11] [B1] [B2] [B3] [B3] [B4] [B5] [B6] [B7]	Credit / Debit card	VIP: 17*788.90 = 13411.30	13411.30
2	15 14 10	30	VIP: [A15] [A1] [B1] [C1] [C2] [C3] [C4] [C5] [C6] [C7] [C8] [C9] [C11] [D1] [C10] General: [A1] [A2] [A3] [A4] [A5] [B1] [B2] [B3][B4] [B5] [A16] [A17] [A18] [A19] [A20] [B16] [B17] [B18] [B19] [B20] [A6] [A7] [A8] [A9] [A10] [A11] [A12] [A13] [A14] [A15]	Credit / Debit card	VIP: 10*788.90 = 7889.00 General: 30*259.90 = 7797.00	7889.00 + 7797.00 = 15686.00
3	*5* *2*	*5*	*VIP: [B9] [B10]* *General: [C1] [C2] [C3] [C4][C5]*	*N/A (cancel order)*	*VIP: 2*788.90 = 1577.80* *General:* *5*259.90 = 1299.50*	*1577.80 + 1299.50 = 2877.30*

Actual Output

Order 1 (Test for buying seats out of range - A11, seats currently chosen - B3) :

```
----------------------------------------
            Ticket ordering
----------------------------------------


   No.    Ticket Type          Price
----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 17




  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]
```

```
Please enter seat code 1: A1
Please enter seat code 2: A2
Please enter seat code 3: A3
Please enter seat code 4: A4
Please enter seat code 5: A5
Please enter seat code 6: A6
Please enter seat code 7: A7
Please enter seat code 8: A8
Please enter seat code 9: A9
Please enter seat code 10: A10
Please enter seat code 11: A11
Invalid seat code, please enter again: B1
Please enter seat code 12: B2
Please enter seat code 13: B3
Please enter seat code 14: B3
Seat not available, choose another: B4
Please enter seat code 15: B5
Please enter seat code 16: B6
Please enter seat code 17: B7
Do you want add ticket? (Y/N): N
```

//Prompt error message when user input invalid range of seat code (A11) and the seat that is already chosen (B3).

```
  Ticket Category        Price/ticket       Quantity         Amount
*********************************************************************
             VIP              788.90             17       13411.30
---------------------------------------------------------------------
                  Total payment: RM 13411.30
---------------------------------------------------------------------


                  Confirm Payment? (Y/N): Y


      ----------------------------------------
                   Payment Section
      ----------------------------------------


                1 -- Credit/Debit Card
                2 -- Cash
                3 -- E-Wallet

        Choose a payment method: 1
```

```
       Enter credit card number (no space): 1234567890123456

       Expiry Date (MM): 12
       Expiry Date (MM/YY): 12/26

       Enter name on card: Villian Tan


          Transaction successful.


          Print receipt? (Y/N): Y
```

```
 ---------------------------------------------------------------------
|                                                                     |
|           +----------------------------------------+                |
|           |    _____          __  ___  __          |                |
|           |   / ___/__  ___  / / / _ \/ /__ ___ __  |                |
|           |  / /__/ _ \/ _ \/ / / ___/ / _ `/ // /  |                |
|           |  \___/\___/\___/_/ /_/  /_/\_,_/\_, /   |                |
|           |                                 /___/   |                |
|           +----------------------------------------+                |
|                        ** RECEIPT **                                |
|                                                                     |
|               Invoice number  : INV0001                             |
|               Date            : 17/12/2023                          |
|   ==========================================================        |
|                                                                     |
|   Ticket category     Qty       Price (RM)     Amount               |
|   VIP                 17        788.90         13411.30             |
|                                                                     |
|               Seat(s) code    :                                     |
|                                 VA1                                 |
|                                 VA2                                 |
|                                 VA3                                 |
|                                 VA4                                 |
|                                 VA5                                 |
|                                 VA6                                 |
|                                 VA7                                 |
|                                 VA8                                 |
|                                 VA9                                 |
|                                 VA10                                |
|                                 VB1                                 |
|                                 VB2                                 |
|                                 VB3                                 |
|                                 VB4                                 |
|                                 VB5                                 |
|                                 VB6                                 |
|                                 VB7                                 |
|                                                                     |
|   ----------------------------------------------------------        |
|               Total Price     : RM 13411.30                         |
|               Payment Method  : Credit / Debit Card                  |
|               Card Payment Ref: 1234567890123456                     |
|                                                                     |
 ---------------------------------------------------------------------


                  Please take the receipt.
```

```
                                        ----------------------------------------
                                                  Tickets Generated
                                        ----------------------------------------



 ------------------------------------------------------------------------------------------------------------
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                  | Venue :                  |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                        |                          |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                | National Stadium         |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                | Bukit Jalil              |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                        | Kuala Lumpur             |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                        |                          |
| GATE  15A              |                                                                                  |                          |
| SEAT  VA1              |                          >> COOLPLAY WORLD TOUR <<                               | Wednesday                |
|                        |                      _____                          _____                        | December 22, 2023        |
|                        |                    ( ___ )----------------------( ___ )                        |                          |
| GATE Open 6.30pm       |                     |   |  _    _   ___   ____   |   |                         |   ------------------   |
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                         |   | " All MUSIC      |  |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                         |   |   IS BEAUTIFUL " |  |
| Queue No. 17           |                     |   |   \_/  )_____()_(     |   |                         |    ------------------   |
|                        |                     |___|                        |___|                         |                          |
|                        |                   (_____)----------------------(_____)                        | C  O  O  L  P  L  A  Y   |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------


 ------------------------------------------------------------------------------------------------------------
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                  | Venue :                  |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                        |                          |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                | National Stadium         |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                | Bukit Jalil              |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                        | Kuala Lumpur             |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                        |                          |
| GATE  15A              |                                                                                  |                          |
| SEAT  VA2              |                          >> COOLPLAY WORLD TOUR <<                               | Wednesday                |
|                        |                      _____                          _____                        | December 22, 2023        |
|                        |                    ( ___ )----------------------( ___ )                        |                          |
| GATE Open 6.30pm       |                     |   |  _    _   ___   ____   |   |                         |   ------------------   |
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                         |   | " All MUSIC      |  |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                         |   |   IS BEAUTIFUL " |  |
| Queue No. 16           |                     |   |   \_/  )_____()_(     |   |                         |    ------------------   |
|                        |                     |___|                        |___|                         |                          |
|                        |                   (_____)----------------------(_____)                        | C  O  O  L  P  L  A  Y   |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------


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|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                  | Venue :                  |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                        |                          |
|                        |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                | National Stadium         |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                | Bukit Jalil              |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                        | Kuala Lumpur             |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                        |                          |
| GATE  15A              |                                                                                  |                          |
| SEAT  VA3              |                          >> COOLPLAY WORLD TOUR <<                               | Wednesday                |
|                        |                      _____                          _____                        | December 22, 2023        |
|                        |                    ( ___ )----------------------( ___ )                        |                          |
| GATE Open 6.30pm       |                     |   |  _    _   ___   ____   |   |                         |   ------------------   |
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                         |   | " All MUSIC      |  |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                         |   |   IS BEAUTIFUL " |  |
| Queue No. 15           |                     |   |   \_/  )_____()_(     |   |                         |    ------------------   |
|                        |                     |___|                        |___|                         |                          |
|                        |                   (_____)----------------------(_____)                        | C  O  O  L  P  L  A  Y   |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------


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|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                  | Venue :                  |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                        |                          |
```

```
|                       |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)       | National Stadium        |
| **FREE T-SHIRT        |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|       | Bukit Jalil             |
| **FREE GLOW STICK     |   \__ \  __ / __ | _|     /  _ | \__ \                                | Kuala Lumpur            |
|                       |   ____/ _|  _| _| ___| _|_\ ___| ____/                                |                         |
| GATE  15A             |                                                                         |                         |
| SEAT  VA4             |                        >> COOLPLAY WORLD TOUR <<                        | Wednesday               |
|                       |                    _____                          _____                 | December 22, 2023       |
|                       |                  ( ___ )----------------------( ___ )                  |                         |
| GATE Open 6.30pm      |                   |   |  _   _   ___   ____   |   |                    |   _________________   |
|                       |                   |   | \ ( ) / )_ _( )  _)\  |   |                    |  | " All MUSIC      |  |
|                       |                   |   |  )\_/(  _| |_ | '__/  |   |                    |  |   IS BEAUTIFUL " |  |
| Queue No. 14          |                   |   |   \_/  )_____()_(     |   |                    |   -----------------    |
|                       |                   |___|                       |___|                    |                         |
|                       |                  (_____)----------------------(_____)                  | C  O  O  L  P  L  A  Y  |
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ---------------------------------------------------------------------------------------------------------------------------------------
```

```
 ---------------------------------------------------------------------------------------------------------------------------------------
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                   |                                                                         | Venue :                 |
| RM    788.90          |     \  |  |  |   __| _ _|   __|           _|    |    |                |                         |
|                       |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)       | National Stadium        |
| **FREE T-SHIRT        |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|       | Bukit Jalil             |
| **FREE GLOW STICK     |   \__ \  __ / __ | _|     /  _ | \__ \                                | Kuala Lumpur            |
|                       |   ____/ _|  _| _| ___| _|_\ ___| ____/                                |                         |
| GATE  15A             |                                                                         |                         |
| SEAT  VA5             |                        >> COOLPLAY WORLD TOUR <<                        | Wednesday               |
|                       |                    _____                          _____                 | December 22, 2023       |
|                       |                  ( ___ )----------------------( ___ )                  |                         |
| GATE Open 6.30pm      |                   |   |  _   _   ___   ____   |   |                    |   _________________   |
|                       |                   |   | \ ( ) / )_ _( )  _)\  |   |                    |  | " All MUSIC      |  |
|                       |                   |   |  )\_/(  _| |_ | '__/  |   |                    |  |   IS BEAUTIFUL " |  |
| Queue No. 13          |                   |   |   \_/  )_____()_(     |   |                    |   -----------------    |
|                       |                   |___|                       |___|                    |                         |
|                       |                  (_____)----------------------(_____)                  | C  O  O  L  P  L  A  Y  |
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ---------------------------------------------------------------------------------------------------------------------------------------
```

-
-
-
- / / tickets generated

```
 ---------------------------------------------------------------------------------------------------------------------------------------
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                   |                                                                         | Venue :                 |
| RM    788.90          |     \  |  |  |   __| _ _|   __|           _|    |    |                |                         |
|                       |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)       | National Stadium        |
| **FREE T-SHIRT        |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|       | Bukit Jalil             |
| **FREE GLOW STICK     |   \__ \  __ / __ | _|     /  _ | \__ \                                | Kuala Lumpur            |
|                       |   ____/ _|  _| _| ___| _|_\ ___| ____/                                |                         |
| GATE  15A             |                                                                         |                         |
| SEAT  VB7             |                        >> COOLPLAY WORLD TOUR <<                        | Wednesday               |
|                       |                    _____                          _____                 | December 22, 2023       |
|                       |                  ( ___ )----------------------( ___ )                  |                         |
| GATE Open 6.30pm      |                   |   |  _   _   ___   ____   |   |                    |   _________________   |
|                       |                   |   | \ ( ) / )_ _( )  _)\  |   |                    |  | " All MUSIC      |  |
|                       |                   |   |  )\_/(  _| |_ | '__/  |   |                    |  |   IS BEAUTIFUL " |  |
| Queue No. 1           |                   |   |   \_/  )_____()_(     |   |                    |   -----------------    |
|                       |                   |___|                       |___|                    |                         |
|                       |                  (_____)----------------------(_____)                  | C  O  O  L  P  L  A  Y  |
|                       | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ---------------------------------------------------------------------------------------------------------------------------------------


                                Press [Y] if done ... ...|
```

Order 2 (Test for buying seats out of range - A15, C11, D1, seats reserved - A1, B1 and buying seats more than available seats) :

```
----------------------------------------
            Ticket ordering
----------------------------------------


   No.    Ticket Type          Price
----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 15
Sorry, not enough ticket. Please reduce your amount.
There are 13 seat(s) left.
Please enter amount of ticket to buy: 14
Sorry, not enough ticket. Please reduce your amount.
There are 13 seat(s) left.
Please enter amount of ticket to buy: 10




  [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
  [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
  [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]
```

//Tell user there is not enough tickets, ask to reduce the number of seats

```
Please enter seat code 1: A15
Seat not available, choose another: A1
Seat not available, choose another: B1
Seat not available, choose another: C1
Please enter seat code 2: C2
Please enter seat code 3: C3
Please enter seat code 4: C4
Please enter seat code 5: C5
Please enter seat code 6: C6
Please enter seat code 7: C7
Please enter seat code 8: C8
Please enter seat code 9: C9
Please enter seat code 10: C11
Invalid seat code, please enter again: D1
Invalid seat code, please enter again: C10
Do you want add ticket? (Y/N): Y
```

```
                    ----------------------------------------
                                Ticket ordering
                    ----------------------------------------


                       No.    Ticket Type          Price
                    ----------------------------------------
                         1            VIP         788.90
                         2        General         259.90


                   Please enter the category of ticket: 2


                   Please enter amount of ticket to buy: 30



[A1][A2][A3][A4][A5]   [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15]   [A16][A17][A18][A19][A20]
[B1][B2][B3][B4][B5]   [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15]   [B16][B17][B18][B19][B20]
[C1][C2][C3][C4][C5]   [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15]   [C16][C17][C18][C19][C20]
[D1][D2][D3][D4][D5]   [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15]   [D16][D17][D18][D19][D20]
[E1][E2][E3][E4][E5]   [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15]   [E16][E17][E18][E19][E20]
[F1][F2][F3][F4][F5]   [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15]   [F16][F17][F18][F19][F20]
[G1][G2][G3][G4][G5]   [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15]   [G16][G17][G18][G19][G20]
```

```
Please enter seat code 1: A1
Please enter seat code 2: A2
Please enter seat code 3: A3
Please enter seat code 4: A4
Please enter seat code 5: A5
Please enter seat code 6: B1
Please enter seat code 7: B2
Please enter seat code 8: B3
Please enter seat code 9: B4
Please enter seat code 10: B5
Please enter seat code 11: A16
Please enter seat code 12: A17
Please enter seat code 13: A18
Please enter seat code 14: A19
Please enter seat code 15: A20
Please enter seat code 16: B16
Please enter seat code 17: B17
Please enter seat code 18: B18
Please enter seat code 19: B19
Please enter seat code 20: B20
Please enter seat code 21: A6
Please enter seat code 22: A7
Please enter seat code 23: A8
Please enter seat code 24: A9
Please enter seat code 25: A10
Please enter seat code 26: A11
Please enter seat code 27: A12
Please enter seat code 28: A13
Please enter seat code 29: A14
Please enter seat code 30: A15
Do you want add ticket? (Y/N): N
```

```
  Ticket Category        Price/ticket       Quantity         Amount
 ******************************************************************
              VIP              788.90             10        7889.00
          General              259.90             30        7797.00
-------------------------------------------------------------------
                  Total payment: RM 15686.00
-------------------------------------------------------------------


                  Confirm Payment? (Y/N): Y


      ----------------------------------------
                   Payment Section
      ----------------------------------------


                1 -- Credit/Debit Card
                2 -- Cash
                3 -- E-Wallet

        Choose a payment method: 1

        Enter credit card number (no space): 1234567890123456

        Expiry Date (MM): 4
        Expiry Date (MM/YY): 4/27

        Enter name on card: Chong


           Transaction successful.


           Print receipt? (Y/N): y
```

```
 ----------------------------------------------------------------
|                                                                |
|           +----------------------------------------+           |
|           |    _____          __  ___  __          |           |
|           |   / ___/__  ___  / / / _ \/ /__ ___ __ |           |
|           |  / /__/ _ \/ _ \/ / / ___/ / _ `/ // / |           |
|           |  \___/\___/\___/_/ /_/  /_/\_,_/\_, /  |           |
|           |                                /___/   |           |
|           +----------------------------------------+           |
|                        ** RECEIPT **                           |
|                                                                |
|               Invoice number  : INV0002                        |
|               Date            : 17/12/2023                     |
|   ==========================================================   |
|                                                                |
|   Ticket category     Qty       Price (RM)     Amount          |
|   VIP                 10        788.90         7889.00         |
|   General             30        259.90         7797.00         |
|                                                                |
|               Seat(s) code    :                                |
|                                 VC1                            |
|                                 VC2                            |
|                                 VC3                            |
|                                 VC4                            |
|                                 VC5                            |
|                                 VC6                            |
|                                 VC7                            |
|                                 VC8                            |
|                                 VC9                            |
|                                 VC10                           |
|                                 A1                             |
|                                 A2                             |
|                                 A3                             |
|                                 A4                             |
|                                 A5                             |
|                                 A6                             |
|                                 A7                             |
|                                 A8                             |
```

```
|                                A9                              |
|                                A10                             |
|                                A11                             |
|                                A12                             |
|                                A13                             |
|                                A14                             |
|                                A15                             |
|                                A16                             |
|                                A17                             |
|                                A18                             |
|                                A19                             |
|                                A20                             |
|                                B1                              |
|                                B2                              |
|                                B3                              |
|                                B4                              |
|                                B5                              |
|                                B16                             |
|                                B17                             |
|                                B18                             |
|                                B19                             |
|                                B20                             |
|                                                                |
|   ----------------------------------------------------------   |
|              Total Price     : RM 15686.00                     |
|              Payment Method  : Credit / Debit Card              |
|              Card Payment Ref: 1234567890123456                 |
|                                                                |
 ----------------------------------------------------------------
```

```
                                    ----------------------------------------
                                              Tickets Generated
                                    ----------------------------------------


 ------------------------------------------------------------------------------------------------------------------------
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            | Venue :                 |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                         |
|                        |    |\/ |  |  | \__ \   |   (         _ \   _|     _|    \    -_)                        | National Stadium        |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil             |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                  | Kuala Lumpur            |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                  |                         |
| GATE  15A              |                                                                                            |                         |
| SEAT  VC1              |                          >> COOLPLAY WORLD TOUR <<                                         | Wednesday               |
|                        |                       _____                        _____                                  | December 22, 2023       |
|                        |                    ( ___ )----------------------( ___ )                                  |                         |
| GATE Open 6.30pm       |                     |   |  _   _   ___   ____   |   |                                    |   _________________   |
|                        |                     |   | \ ( ) / )_ _( )  _)\  |   |                                    |  | " All MUSIC      |  |
|                        |                     |   |  )\_/(  _| |_ | '__/  |   |                                    |  |   IS BEAUTIFUL " |  |
| Queue No. 57           |                     |   |   \_/  )_____()_(     |   |                                    |   -----------------   |
|                        |                     |___|                       |___|                                    |                         |
|                        |                    (_____)----------------------(_____)                                  | C  O  O  L  P  L  A  Y  |
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ------------------------------------------------------------------------------------------------------------------------


 ------------------------------------------------------------------------------------------------------------------------
|                        | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                    |                                                                                            | Venue :                 |
| RM    788.90           |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                         |
|                        |    |\/ |  |  | \__ \   |   (         _ \   _|     _|    \    -_)                        | National Stadium        |
| **FREE T-SHIRT         |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil             |
| **FREE GLOW STICK      |   \__ \  __ / __ | _|     /  _ | \__ \                                                  | Kuala Lumpur            |
|                        |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                  |                         |
| GATE  15A              |                                                                                            |                         |
```

```
| SEAT  VC2            |                        >> COOLPLAY WORLD TOUR <<                    | Wednesday                |
|                      |                     _____                  _____                    | December 22, 2023        |
|                      |                  ( ___ )----------------------( ___ )               |                          |
| GATE Open 6.30pm     |                   |   |  _   _   ___   ____   |   |                |   ------------------   |
|                      |                   |   | \ ( ) / )_ _( )  _)\  |   |                |   | " All MUSIC      |
|                      |                   |   |  )\_/(  _| |_ | '__/  |   |                |   |   IS BEAUTIFUL " |
| Queue No. 56         |                   |   |   \_/  )_____()_(     |   |                |   ------------------   |
|                      |                   |___|                       |___|                |                          |
|                      |                 (_____)----------------------(_____)               | C  O  O  L  P  L  A  Y  |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ----------------------------------------------------------------------------------------------------------------------------------


 ----------------------------------------------------------------------------------------------------------------------------------
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
| VIP                  |                                                                                                                  | Venue :                  |
| RM    788.90         |     \  |  |  |   __| _ _|   __|           _|    |    |                                  |                          |
|                      |    |\/ |  |  | \__ \   |   (        _ \   _|     _|    \    -_)                        | National Stadium         |
| **FREE T-SHIRT       |   _|__|| \_\/| _|__/_|__| \___||  \__|/ _|     \__| _| _| \___|                        | Bukit Jalil              |
| **FREE GLOW STICK    |   \__ \  __ / __ | _|     /  _ | \__ \                                                  | Kuala Lumpur             |
|                      |   ____/ _|  _| _| ___| _|_\ ___| ____/                                                   |                          |
| GATE  15A            |                                                                                                                  |                          |
| SEAT  VC3            |                        >> COOLPLAY WORLD TOUR <<                                                 | Wednesday                |
|                      |                     _____                  _____                                                 | December 22, 2023        |
|                      |                  ( ___ )----------------------( ___ )                                            |                          |
| GATE Open 6.30pm     |                   |   |  _   _   ___   ____   |   |                                             |   ------------------   |
|                      |                   |   | \ ( ) / )_ _( )  _)\  |   |                                             |   | " All MUSIC      |
|                      |                   |   |  )\_/(  _| |_ | '__/  |   |                                             |   |   IS BEAUTIFUL " |
| Queue No. 55         |                   |   |   \_/  )_____()_(     |   |                                             |   ------------------   |
|                      |                   |___|                       |___|                                             |                          |
|                      |                 (_____)----------------------(_____)                                            | C  O  O  L  P  L  A  Y  |
|                      | *VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP* |
 ----------------------------------------------------------------------------------------------------------------------------------
```

-
-
-

. tickets generated

```
| RM    259.90         |    __  __              _                  __   _    _                                  |                          |
|                      |   |  \/  |            (_)                / _| | |  | |                                 | National Stadium         |
|                      |   | \  / | _   _  ___  _   ___    ___  | |_  | |_ | |__    ___                        | Bukit Jalil              |
|                      |   | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                       | Kuala Lumpur             |
|                      |   | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/                        |                          |
| GATE  15A            |   |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|                        |                          |
| SEAT  B19            |                                                                                                                  | Wednesday                |
|                      |                 _____          _                                                          | December 22, 2023        |
|                      |                / ____|        | |                                                         |                          |
| GATE Open 7.00pm     |               | (___   _ __  | |__    ___  _ __  ___  ___                                |   ------------------   |
|                      |                \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|                                |   | " WHERE MEMORIES |
|                      |                ____) || |_) || | | ||  __/| |   |  __/\__ \                                |   |    ARE CREATED " |
| Queue No. 19         |               |_____/ | .__/ |_| |_| \___||_|   \___||___/                                |   ------------------   |
|                      |                      | |                                                                |                          |
|                      |                      |_|                                                                | C  O  O  L  P  L  A  Y  |
 ----------------------------------------------------------------------------------------------------------------------------------


 ----------------------------------------------------------------------------------------------------------------------------------
| General              |                          **COOLPLAY WORLD TOUR**                                                 | Venue :                  |
| RM    259.90         |    __  __              _                  __   _    _                                  |                          |
|                      |   |  \/  |            (_)                / _| | |  | |                                 | National Stadium         |
|                      |   | \  / | _   _  ___  _   ___    ___  | |_  | |_ | |__    ___                        | Bukit Jalil              |
|                      |   | |\/| || | | |/ __|| | / __|  / _ \ |  _| | __|| '_ \  / _ \                       | Kuala Lumpur             |
|                      |   | |  | || |_| |\__ \| || (__  | (_) || |   | |_ | | | ||  __/                        |                          |
| GATE  15A            |   |_|  |_| \__,_||___/|_| \___|  \___/ |_|    \__||_| |_| \___|                        |                          |
| SEAT  B20            |                                                                                                                  | Wednesday                |
|                      |                 _____          _                                                          | December 22, 2023        |
|                      |                / ____|        | |                                                         |                          |
| GATE Open 7.00pm     |               | (___   _ __  | |__    ___  _ __  ___  ___                                |   ------------------   |
|                      |                \___ \ | '_ \ | '_ \  / _ \| '__|/ _ \/ __|                                |   | " WHERE MEMORIES |
|                      |                ____) || |_) || | | ||  __/| |   |  __/\__ \                                |   |    ARE CREATED " |
| Queue No. 18         |               |_____/ | .__/ |_| |_| \___||_|   \___||___/                                |   ------------------   |
|                      |                      | |                                                                |                          |
|                      |                      |_|                                                                | C  O  O  L  P  L  A  Y  |
 ----------------------------------------------------------------------------------------------------------------------------------


                                Press [Y] if done ... ...y
```

Order 3 (Cancel Order) :

```
-----------------------------------------
            Ticket ordering
-----------------------------------------


   No.    Ticket Type          Price
-----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 5
Sorry, not enough ticket. Please reduce your amount.
There are 3 seat(s) left.
Please enter amount of ticket to buy: 2




 [A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
 [B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
 [C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]


        Please enter seat code 1: B9
        Please enter seat code 2: B10
        Do you want add ticket? (Y/N): Y
```

```
                         -----------------------------------------
                                    Ticket ordering
                         -----------------------------------------


                            No.    Ticket Type          Price
                         -----------------------------------------
                              1            VIP         788.90
                              2        General         259.90


                        Please enter the category of ticket: 2


                        Please enter amount of ticket to buy: 5



 [A1][A2][A3][A4][A5]   [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15]   [A16][A17][A18][A19][A20]
 [B1][B2][B3][B4][B5]   [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15]   [B16][B17][B18][B19][B20]
 [C1][C2][C3][C4][C5]   [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15]   [C16][C17][C18][C19][C20]
 [D1][D2][D3][D4][D5]   [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15]   [D16][D17][D18][D19][D20]
 [E1][E2][E3][E4][E5]   [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15]   [E16][E17][E18][E19][E20]
 [F1][F2][F3][F4][F5]   [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15]   [F16][F17][F18][F19][F20]
 [G1][G2][G3][G4][G5]   [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15]   [G16][G17][G18][G19][G20]


                                Please enter seat code 1: C1
                                Please enter seat code 2: C2
                                Please enter seat code 3: C3
                                Please enter seat code 4: C4
                                Please enter seat code 5: C5
                                Do you want add ticket? (Y/N): N
```

```
  Ticket Category        Price/ticket       Quantity         Amount
*********************************************************************
             VIP              788.90              2        1577.80
         General              259.90              5        1299.50
---------------------------------------------------------------------
                 Total payment: RM 2877.30
---------------------------------------------------------------------


                 Confirm Payment? (Y/N): N
```

//No confirm payment (cancel order)

Sales Report

Order No.	Expected output (after customer order)			
	Category	**Seat left**	**Seat Sold**	**Profit (RM)**
1	VIP	13	17	13411.30
	General	140	0	0.00
	Total Profit: 13411.30 No. of customer: 1 VIP bar (dots): 17 General bar (dots): 0			
2	VIP	3	27	21300.30
	General	110	30	7797.00
	Total Profit: 29097.30 No. of customer: 2 VIP bar (dots): 27 General bar (dots): 30			
3 (no changes for cancelling order)	VIP	3	27	21300.30
	General	110	30	7797.00
	Total Profit: 29097.30 No. of customer: 2 VIP bar (dots): 27 General bar (dots): 30			

Actual Output of Final Sales Report:

```
==============================================================================

            ____        _            ____                        _
           / ___|  __ _| | ___  ___  |  _ \ ___ _ __   ___  _ __| |_
           \___ \ / _` | |/ _ \/ __| | |_) / _ \ '_ \ / _ \| '__| __|
            ___) | (_| | |  __/\__ \ |  _ <  __/ |_) | (_) | |  | |_
           |____/ \__,_|_|\___||___/ |_| \_\___| .__/ \___/|_|   \__|
                                                |_|
          Date: 17 December 2023
          Event: CoolPlay Concert
          Sales Person: user1
------------------------------------------------------------------------------

                     +=======================================+
                     |____  __________  ___  _______  _______|
                     ||  |  |__|__|__|\ ||   [__|__|  |__[__ |
                     ||__|__| \| \|__| \||   ___|  |__|_____]|
                     +=======================================+

   TypeSeat      | PRICE (RM) | TOTAL SEAT | SEAT LEFT | SEAT SOLD | PROFIT (RM)
-----------------|------------|------------|-----------|-----------|------------
      VIP        |     788.90 |         30 |         3|         27|   21300.30
    GENERAL      |     250.90 |        140 |       110|         30|    7797.00
-------------------------------------------------------------------------------
  TOTAL PROFIT(RM)                                                    29097.30
-------------------------------------------------------------------------------
 Number of customer(s): 2

                            *
                            *
                            *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
```

```
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                            *               *
                        General             VIP

 ==============================================================================
 Note that one * = 1 tickets


                      Press [Y] back to menu : |
```

Seats Display

Order No.	Option	Expected Output

1	2 (VIP)	```
		---------- 	
		VIP Seats Layout
		---------- 	
		
		[A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
		[B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
		[C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]
		``
		//no changes after cancel order

2	3 (Overall)	```
		---------- 	
		| |
		| Stage |
		| |
		---------- 	
		
		[A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
		[B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
		[C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]
		
		[A1][A2][A3][A4][A5] [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15] [A16][A17][A18][A19][A20]
		[B1][B2][B3][B4][B5] [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15] [B16][B17][B18][B19][B20]
		[C1][C2][C3][C4][C5] [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15] [C16][C17][C18][C19][C20]
		[D1][D2][D3][D4][D5] [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15] [D16][D17][D18][D19][D20]
		[E1][E2][E3][E4][E5] [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15] [E16][E17][E18][E19][E20]
		[F1][F2][F3][F4][F5] [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15] [F16][F17][F18][F19][F20]
		[G1][G2][G3][G4][G5] [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15] [G16][G17][G18][G19][G20]
		``
		//no changes after cancel order

3	3	```
		---------- 	
		| |
		| Stage |
		| |
		---------- 	
		
		[A1][A2][A3][A4][A5][A6][A7][A8][A9][A10]
		[B1][B2][B3][B4][B5][B6][B7][B8][B9][B10]
		[C1][C2][C3][C4][C5][C6][C7][C8][C9][C10]
		
		[A1][A2][A3][A4][A5] [A6][A7][A8][A9][A10][A11][A12][A13][A14][A15] [A16][A17][A18][A19][A20]
		[B1][B2][B3][B4][B5] [B6][B7][B8][B9][B10][B11][B12][B13][B14][B15] [B16][B17][B18][B19][B20]
		[C1][C2][C3][C4][C5] [C6][C7][C8][C9][C10][C11][C12][C13][C14][C15] [C16][C17][C18][C19][C20]
		[D1][D2][D3][D4][D5] [D6][D7][D8][D9][D10][D11][D12][D13][D14][D15] [D16][D17][D18][D19][D20]
		[E1][E2][E3][E4][E5] [E6][E7][E8][E9][E10][E11][E12][E13][E14][E15] [E16][E17][E18][E19][E20]
		[F1][F2][F3][F4][F5] [F6][F7][F8][F9][F10][F11][F12][F13][F14][F15] [F16][F17][F18][F19][F20]
		[G1][G2][G3][G4][G5] [G6][G7][G8][G9][G10][G11][G12][G13][G14][G15] [G16][G17][G18][G19][G20]
		``
		//no changes after cancel order

Extra cases

Case for choosing a ticket category with no more tickets left:

```
-----------------------------------------
            Ticket ordering
-----------------------------------------


   No.    Ticket Type          Price
-----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1
    Sorry, no tickets left for this category.
    Kindly order other category.



    Press [Y] go back to menu ... ...Y
```

Case when all tickets are sold out:

```
-----------------------------------------
            Ticket ordering
-----------------------------------------


   No.    Ticket Type          Price
-----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 1


Please enter amount of ticket to buy: 30
```

```
-----------------------------------------
            Ticket ordering
-----------------------------------------


   No.    Ticket Type          Price
-----------------------------------------
     1            VIP         788.90
     2        General         259.90


Please enter the category of ticket: 2


Please enter amount of ticket to buy: 140
```

//System closed and stop looping back to menu

```
| Queue No. 31          |          |_____/ | .__/ |_| |_| \___||_|   \___||___/              |    ------------------   |
|                       |                 | |                                              |                         |
|                       |                 |_|                                              | C  O  O  L  P  L  A  Y  |
-----------------------------------------------------------------------------------------------------------------------


                                Press [Y] if done ... ...y



                        Thank you for using this ticketing system.
                                   Have a nice day !!
```

Case when choosing menu option [4] Close system to quit program :

```
  ___                    _        ___    _
 (  _`\                 (_ )     (  _`\(_ )
 | ( (_)   _      _     | |     | |_) )| |    _ _  _   _
 | |  _  /'_`\  /'_`\   | |     | ,__/'| |  /'_` )( ) ( )
 | (_( )( (_) )( (_) )  | |     | |     | | ( (_| || (_) |
 (____/''\___/''\___/'(___)    (_)    (___)`\__,_)`\__, |
                                                    ( )_| |
                                                    `\___/'


    Welcome to concert ticketing system for
   ----------------------------------------
               Cool Play Concert
   ----------------------------------------


              1 -- Seat display
              2 -- Order ticket
              3 -- Sales report
              4 -- Close system

             Select an option: 4
```

```
        Thank you for using this ticketing system.
                    Have a nice day !!



ignment\x64\Debug\Assignment.exe (process 14704) exited with code 0.
e console when debugging stops, enable Tools->Options->Debugging->Aut
is window . . .
```

Program Listing

```
#include <iostream>
#include <iomanip>
#include <windows.h>
#include <string>
#include <cctype>
#include <cstring>
#include <cstdlib>
#include <cmath>
#define RED_TEXT "\033[1;31m"  //Code for red color output
#define TOTAL_TICKETS 170
using namespace std;

void login();
void printTitle();
void menu(bool* onGoing);
void seatDisplay();
void vipSeats();
void generalSeats();
void orderTicket();
double calculate();
void payment(double total, string* receiptMsg, string* payMethod);
void receipt(string* receiptMsg, string* payMethod);
void ticket();
bool validateQuantity(char category, int* seatsQuantity);
bool validateSeats(char category, int* seatsQuantity);
void salesReport();

int tickets_sold = 0;
int cust_num = 0;
const string ticketType[2] = { "VIP", "General" };
const double ticketPrice[2] = { 788.90, 259.90 };
const int VIPROW = 3, VIPCOL = 10;
const int GENROW = 7, GENCOL = 20;
int vipAvailib = VIPROW * VIPCOL;
int generalAvailib = GENROW * GENCOL;
int currentVIPAmount = 0;
int currentGenAmount = 0;
bool currentVIPSeats[VIPROW][VIPCOL] = { false };  //array storing current seats chosen by
user
bool currentGenSeats[GENROW][GENCOL] = { false };
double currentTotal;
bool vipSeatsReserv[VIPROW][VIPCOL] = { false }; //boolean array to indicate seats being
reserved or not
bool genSeatsReserv[GENROW][GENCOL] = { false };

int main()
```

```
{
	bool onGoing = true;
	login();
	do {
		system("CLS");
		currentVIPAmount = 0;
		currentGenAmount = 0;
		currentTotal = 0;				//clear all data for new customer
		for (int row = 0; row < VIPROW; ++row) {
			for (int col = 0; col < VIPCOL; ++col) {
				currentVIPSeats[row][col] = false;		//initialize all seats
chosen to empty (false)
			};
		};
		for (int row = 0; row < GENROW; ++row) {
			for (int col = 0; col < GENCOL; ++col) {
				currentGenSeats[row][col] = false;
			};
		};
		menu(&onGoing);
		cout << "\n\n\n";
	} while (tickets_sold < TOTAL_TICKETS && onGoing); //continue loop system until
sold out or being quit

	cout << setw(80) << "Thank you for using this ticketing system." << endl;
	cout << setw(68) << "Have a nice day !!" << endl;
	cout << "\n\n\n";

	return 0;
}

void login() {
	printTitle();
	string username = "user1";
	string password = "password";
	string usernameInput, pswInput;
	cout << setw(77) << "Cool Play Concert Ticketing System" << endl;
	cout << setw(80) << "========================================" << endl;
	cout << setw(64) << "Login Page" << endl;
	cout << setw(80) << "========================================" << endl;
	cout << "\n\n\n";

	do {
		cout << setw(65) << "Enter username: ";
		getline(cin, usernameInput);
		if (username != usernameInput) {
			cout << setw(77) << "Invalid username. Please reenter." << endl;
		}
```

```
	} while (username != usernameInput);		//validation for correct
username
	do {
		cout << setw(65) << "Enter password: ";
		getline(cin, pswInput);
		if (pswInput != password) {
			cout << setw(70) << "Wrong password. Try again." << endl;
		}
	} while (pswInput != password);		//validation for
correct password
}

void printTitle() {
	cout << "\n\n";
	cout << setw(90) << " ___                _      ___   _                 " << endl;
	cout << setw(90) << "(  _`\\              (_ )    (  _`\\(_ )              " << endl;
	cout << setw(90) << "| ( (_)   _      _    | |    | |_) )| |    _ _  _   _ " << endl;
	cout << setw(90) << "| |  _  /'_`\\  /'_`\\  | |    | ,__/'| |  /'_` )( ) ( )" << endl;
	cout << setw(90) << "| (_( )( (_) )( (_) ) | |    | |    | | ( (_| || (_) |" << endl;
	cout << setw(90) << "(_____/'`\\___/'`\\___/'(___)   (_)   (___)`\\__,_)`\\__, |" << endl;
	cout << setw(90) << "                                                  ( )_| |" << endl;
	cout << setw(90) << "                                                  `\\___/'" << endl;
	cout << "\n\n";
}

void menu(bool* onGoing) {
	char menuOpt;

	printTitle();
	cout << setw(80) << "Welcome to concert ticketing system for " << endl;
	cout << setw(80) << "----------------------------------------" << endl;
	cout << setw(68) << "Cool Play Concert" << endl;
	cout << setw(80) << "----------------------------------------" << endl;
	cout << "\n\n";
	cout << setw(67) << "1 -- Seat display" << endl;
	cout << setw(67) << "2 -- Order ticket" << endl;
	cout << setw(67) << "3 -- Sales report" << endl;
	cout << setw(67) << "4 -- Close system" << endl;
	cout << "\n";
	cout << setw(67) << "Select an option: ";
	cin >> menuOpt;

	while (menuOpt != '1' && menuOpt != '2' && menuOpt != '3' && menuOpt != '4') {
		cout << setw(80) << "Invalid option, please enter again (1/2/3/4): ";
		cin >> menuOpt;
	}		//validation
for correct menu option
	switch (menuOpt) {
```

```
        case '1':
                system("CLS");
                seatDisplay();
                break;
        case '2':
                system("CLS");
                orderTicket();
                break;
        case '3':
                system("CLS");
                salesReport();
                break;
        case '4':
                system("CLS");
                *onGoing = false;       //quit program
        };
        cout << "\n\n\n";
}

void seatDisplay() {
        char category;
        char action;

        cout << "\n\n";
        cout << setw(90) << "   ________      __                          " << endl;
        cout << setw(90) << " /   ------.   / ._`_                        " << endl;
        cout << setw(90) << "|  /         ~--~    \\                       " << endl;
        cout << setw(90) << "| |             __     `._____________________ _^-----^" << endl;
        cout << setw(90) << "| |  I=|=======/--\\=========================| o o o |" <<
endl;
        cout << setw(90) << "\\ |  I=|=======\\__/=========================|_o_o_o_|"
<< endl;
        cout << setw(90) << " \\|                  /                    ~    ~ " << endl;
        cout << setw(90) << "   \\       .---.    .                          " << endl;
        cout << setw(90) << "     -----'     ~~"                            " << endl << "\n\n";

        cout << setw(80) << "----------------------------------------" << endl;
        cout << setw(67) << "Seats Display" << endl;
        cout << setw(80) << "----------------------------------------" << endl << "\n";
        cout << setw(68) << "1 -- General Seats" << endl;
        cout << setw(64) << "2 -- VIP Seats" << endl;
        cout << setw(69) << "3 -- Overall Layout" << endl << "\n";
        cout << setw(65) << "Choose a category: ";
        cin >> category;

        while (category != '1' && category != '2' && category != '3') {
                cout << setw(82) << "Invalid input. Please enter again (1/2/3): ";
                cin >> category;
```

```
        }                                                   //validation for correct category
option

        if (category == '1') {
                cout << "\n";
                cout << setw(80) << "----------------------------------------" << endl;
                cout << setw(70) << "General Seats Layout" << endl;
                cout << setw(80) << "----------------------------------------" << endl;
                generalSeats();
        }
        else if (category == '2') {
                cout << "\n";
                cout << setw(80) << "----------------------------------------" << endl;
                cout << setw(66) << "VIP Seats Layout" << endl;
                cout << setw(80) << "----------------------------------------" << endl;
                vipSeats();
        }
        else {
                cout << "\n\n";
                cout << setw(91) << "-----------------------------------------------------------" << endl;
                cout << setw(30) << "|" << setw(62) << "|" << endl;
                cout << setw(30) << "|" << setw(62) << "|" << endl;
                cout << setw(30) << "|" << setw(32) << "Stage" << setw(30) << "|" << endl;
                cout << setw(30) << "|" << setw(62) << "|" << endl;
                cout << setw(30) << "|" << setw(62) << "|" << endl;
                cout << setw(91) << "-----------------------------------------------------------" << endl;
                vipSeats();
                generalSeats();
        }
        cout << "\n\n";
        cout << setw(67) << "1 -- Back to menu" << endl;
        cout << setw(71) << "2 -- Proceed to order" << endl;
        cout << setw(67) << "Choose an option: ";
        cin >> action;
        while (action != '1' && action != '2') {
                cout << setw(82) << "Invalid input. Please enter again (1/2): ";
                cin >> action;                  //validation for corect action option
        };

        if (action == '2') {
                system("CLS");
                orderTicket();          //proceed to order
        };                                              //else auto loop back to menu() in main()
}


void generalSeats() {
        cout << "\n\n";
```

```
	for (int i = 0; i < GENROW; ++i) {
		cout << "\t\t";
		for (int j = 1; j <= GENCOL; ++j) {
			if (genSeatsReserv[i][j - 1]) {		//if already reserved (true)
				cout << RED_TEXT << "[" << char('A' + i) << j << "]" <<
"\033[0m";  //display red seats
			}
			else {
				cout << "[" << char('A' + i) << j << "]";//else display normal color
			};
			if (j == 5 || j == 15) {
				cout << "   ";			//layout of seats: 5 columns left,
10 middle, 5 right
			}
		};
		cout << endl;
	};
}

void vipSeats() {
	cout << "\n\n";
	for (int i = 0; i < VIPROW; ++i) {
		cout << "\t\t\t\t\t";
		for (int j = 1; j <= VIPCOL; ++j) {
			if (vipSeatsReserv[i][j - 1]) {		//if already reserved (true)
				cout << RED_TEXT << "[" << char('A' + i) << j << "]" <<
"\033[0m";
			}
			else {
				cout << "[" << char('A' + i) << j << "]";
			};
		};
		cout << endl;
	};
}

void orderTicket() {
	char category;
	char confirm;
	char repeat = 'Y';
	char back = 'N';
	int seatsQuantity;
	string payMethod = "";
	string receiptMsg = "";

	do {
		system("CLS");
		cout << "\n\n";
```

```
		cout << setw(80) << "----------------------------------------" << endl;
		cout << setw(66) << "Ticket ordering" << endl;
		cout << setw(80) << "----------------------------------------" << endl << "\n\n";
		cout << setw(45) << "No." << setw(15) << "Ticket Type" << setw(15) <<
"Price" << endl;
		cout << setw(80) << "----------------------------------------" << endl;
		for (int i = 0; i < 2; ++i) {			//display tickets category and respective
price from constant
			cout << setw(45) << i + 1 << setw(15) << ticketType[i] << fixed <<
setprecision(2) << setw(15) << ticketPrice[i] << endl;
		};
		cout << "\n\n";

		cout << setw(75) << "Please enter the category of ticket: ";
		cin >> category;
		while (category != '1' && category != '2') {
			cout << setw(79) << "Invalid input. Please enter again (1/2): ";
			cin >> category;		//validation for category option
		};
		if (category == '1' && vipAvailib == 0 || category == '2' && generalAvailib == 0)
{		//case when all tickets for the chosen category had sold out
			cout << setw(83) << "Sorry, no tickets left for this category." << endl;
			cout << setw(70) << "Kindly order other category." << endl;
			cout << "\n\n\n";
			cout << setw(75) << "Press [Y] go back to menu ... ...";
//bring user back to menu
			cin >> back;
			while (toupper(back) != 'Y') {
				cout << setw(70) << "Enter [Y] only: ";
				cin >> back;
			};
			break;			//end do while
		}

		if (validateQuantity(category, &seatsQuantity) && validateSeats(category,
&seatsQuantity)) { //validation for quantity and seats to buy (both return true)
			cout << setw(78) << "Do you want add ticket? (Y/N): ";
			cin >> repeat;
			while (toupper(repeat) != 'Y' && toupper(repeat) != 'N') {
				cout << setw(72) << "Invalid input. Only Y/N: ";
				cin >> repeat;		//validate input
			};
		}
	} while (toupper(repeat) == 'Y');

	if (toupper(back) != 'Y') {		//if not going back to menu, continue run the
following statements
		system("CLS");
```

```
		currentTotal = calculate();		//calculation for total price to pay
		cout << setw(75) << "Confirm Payment? (Y/N): ";
		cin >> confirm;
		while (toupper(confirm) != 'Y' && toupper(confirm) != 'N') {
			cout << setw(75) << "Invalid input. Only Y/N: ";
			cin >> confirm;		//validate input
		};
		if (toupper(confirm) == 'Y') {
			payment(currentTotal, &receiptMsg, &payMethod);		//proceed
to payment after confirm payment
			receipt(&receiptMsg, &payMethod);
//proceed to print receipt
			tickets_sold += currentVIPAmount;	//update data
			tickets_sold += currentGenAmount;
			cust_num += 1;
			ticket();		//generate ticket(s)
		}
		else {			//case when user does not confirm payment
			for (int row = 0; row < VIPROW; ++row) {		//set the seat
chosen back to false
				for (int col = 0; col < VIPCOL; ++col) {
					if (currentVIPSeats[row][col]) {
						vipSeatsReserv[row][col] = false;
					}
				};
			};
			for (int row = 0; row < GENROW; ++row) {		//set the seat
chosen back to false
				for (int col = 0; col < GENCOL; ++col) {
					if (currentGenSeats[row][col]) {
						genSeatsReserv[row][col] = false;
					}
				};
			};
			vipAvailib += currentVIPAmount;
			generalAvailib += currentGenAmount;
		};
	};
}

bool validateQuantity(char category, int* seatsQuantity) {
	int amount;
	int ticketAvailib;

	if (category == '1') {
		ticketAvailib = vipAvailib;
	}
	else {
```

```
		ticketAvailib = generalAvailib;
	};

	cout << "\n\n";
	do {
		cout << setw(76) << "Please enter amount of ticket to buy: ";
		cin >> amount;
		if (amount > ticketAvailib) {
			cout << setw(90) << "Sorry, not enough ticket. Please reduce your
amount." << endl;
			cout << setw(48) << "There are " << ticketAvailib << " seat(s) left." <<
endl;
		}
		else if (amount <= 0)
		{
			cout << setw(78) << "Invalid input, must be positive integer." << endl;
		};
	} while (amount > ticketAvailib || amount <= 0);		//validation for valid
amount

	if (category == '1') {				//update data
		currentVIPAmount += amount;
		vipAvailib -= amount;
	}
	else {
		currentGenAmount += amount;
		generalAvailib -= amount;
	}
	*seatsQuantity = amount;	//used in validateSeats()
	return true;
}

bool validateSeats(char category, int* seatsQuantity) {
	char chRow;
	int seatRow, seatCol;

	cout << "\n\n";
	if (category == '1') {
		vipSeats();
	}
	else {
		generalSeats();
	};
	cout << "\n\n";

	for (int i = 1; i <= *seatsQuantity; i++) {		//looping times based on quantity
of seats validated
		cout << setw(70) << "Please enter seat code " << i << ": ";
```

```
		cin >> chRow >> seatCol;		//input e.g. A12 , B3,  G20
		seatRow = toupper(chRow) - 'A';		//convert to integer for seats
array handling

		if (category == '1') {
			while (seatRow < 0 || seatRow >= VIPROW || seatCol <= 0 || seatCol
> VIPCOL || vipSeatsReserv[seatRow][seatCol - 1]) {
				if (vipSeatsReserv[seatRow][seatCol - 1]) {		//seats
reserved (true)
					cout << setw(83) << "Seat not available, choose
another: ";
				}
				else {
					cout << setw(86) << "Invalid seat code, please enter
again: ";
				};
				cin >> chRow >> seatCol;
				seatRow = toupper(chRow) - 'A';
			};
			vipSeatsReserv[seatRow][seatCol - 1] = true;		//update
seats currently chosen to seats reserved
			currentVIPSeats[seatRow][seatCol - 1] = true;
		}
		else {
			while (seatRow < 0 || seatRow >= GENROW || seatCol <= 0 || seatCol
> GENCOL || genSeatsReserv[seatRow][seatCol - 1]) {
				if (genSeatsReserv[seatRow][seatCol - 1]) {
					cout << setw(83) << "Seat not available, choose
another: ";
				}
				else {
					cout << setw(86) << "Invalid seat code, please enter
again: ";
				};
				cin >> chRow >> seatCol;
				seatRow = toupper(chRow) - 'A';
			};
			genSeatsReserv[seatRow][seatCol - 1] = true;
			currentGenSeats[seatRow][seatCol - 1] = true;
		};
	};
	*seatsQuantity = 0; //set back to zero in case adding order
	return true;		//validation succeed
}

double calculate() {
	double total;
	cout << "\n\n\n";
```

```
	cout << right << setw(50) << "Ticket Category" << setw(20) << "Price/ticket" <<
setw(15) << "Quantity" << setw(15) << "Amount" << endl;
	cout << setw(101) << "*********************************************************************"
<< endl;
	cout << fixed << setprecision(2);
	if (currentVIPAmount != 0) {
		cout << setw(50) << ticketType[0] << setw(20) << ticketPrice[0] << setw(15)
<< currentVIPAmount << setw(15) << currentVIPAmount * ticketPrice[0] << endl;
	};
	if (currentGenAmount != 0) {
		cout << setw(50) << ticketType[1] << setw(20) << ticketPrice[1] << setw(15)
<< currentGenAmount << setw(15) << currentGenAmount * ticketPrice[1] << endl;
	}
	total = currentVIPAmount * ticketPrice[0] + currentGenAmount * ticketPrice[1];
	cout << setw(101) << "-------------------------------------------------------------------" << endl;
	cout << setw(69) << "Total payment: RM " << fixed << setprecision(2) << total <<
endl;
	cout << setw(101) << "-------------------------------------------------------------------" << endl;
	cout << "\n\n";

	return total;
}

void payment(double total, string* receiptMsg, string* payMethod) {
	char method;
	cout << "\n\n";
	cout << setw(80) << "----------------------------------------" << endl;
	cout << setw(67) << "Payment Section" << endl;
	cout << setw(80) << "----------------------------------------" << endl << "\n\n";
	cout << setw(71) << "1 -- Credit/Debit Card" << endl;
	cout << setw(58) << "2 -- Cash" << endl;
	cout << setw(62) << "3 -- E-Wallet" << endl << "\n";
	cout << setw(66) << "Choose a payment method: ";
	cin >> method;

	while (method != '1' && method != '2' && method != '3') {
		cout << setw(81) << "Invalid input. Please enter again (1/2/3): ";
		cin >> method;				//validation for input option
	};

	if (method == '1') {
		char creditNum[20];
		int mm, yy;
		string name;
		*payMethod = "Credit / Debit Card";
		cout << "\n";
		cout << setw(78) << "Enter credit card number (no space): ";
		cin >> creditNum;
```

```
		while (strlen(creditNum) != 16) {	//validation: only allow 16-digits number
			cout << setw(92) << "Invalid credit card number. Only 16 digits
allowed." << endl;
			cout << setw(78) << "Enter credit card number (no space): ";
			cin >> creditNum;
		}
		cout << "\n";
		cout << setw(59) << "Expiry Date (MM): ";
		cin >> mm;
		while (mm < 1 || mm > 12) {
			cout << setw(77) << "Please enter a valid month (1 - 12)." << endl;
			cout << setw(59) << "Expiry Date (MM): ";
			cin >> mm;
		}
		cout << setw(62) << "Expiry Date (MM/YY): " << mm << "/";
		cin >> yy;
		while (yy < 24 || yy > 28) {
			cout << setw(77) << "Please enter a valid year (24 - 28)." << endl;
			cout << setw(62) << "Expiry Date (MM/YY): " << mm << "/";
			cin >> yy;
		}
		cout << "\n";
		cin.ignore();
		cout << setw(61) << "Enter name on card: ";
		getline(cin, name);

		*receiptMsg = "Card Payment Ref: ";		//add card number to extra
receipt output
		*receiptMsg += creditNum;

		cout << "\n\n";
		cout << setw(67) << "Transaction successful." << endl;
	}
	else if (method == '2') {
		double cashPaid;
		*payMethod = "Cash";
		cout << setw(66) << "Enter cash paid (RM): ";
		cin >> cashPaid;
		while (cashPaid < 0 || cashPaid < total) {		//validation: cannot be
negative / less then total payment
			if (cashPaid < 0) {
				cout << setw(70) << "Please enter a positive value." << endl;
			}
			else {
				cout << setw(63) << "Payment not enough." << endl;
				cout << setw(60) << "Pay more RM" << fixed <<
setprecision(2) << total - cashPaid << endl;
			}
```

```
                        cout << setw(66) << "Enter cash paid (RM): ";
                        cin >> cashPaid;
                }
                if (cashPaid > total) {         //display balance to return if cash paid more
than total
                        cout << setw(60) << "Balance: RM " << fixed << setprecision(2) <<
cashPaid - total << endl;
                }
                cout << "\n\n";
                cout << setw(60) << "Payment succeed." << endl;

                *receiptMsg = "Balance\t: RM ";                         //add balance return to
extra receipt output
                int integerPart = cashPaid - total;             //solve double into appropriate
string
                *receiptMsg += to_string(integerPart);
                *receiptMsg += "." + to_string((int)round((cashPaid - total - integerPart) *
100));
        }
        else {
                *payMethod = "E-Wallet";
                cout << "\n\n";
                cout << setw(66) << "Transaction successful." << endl;
        };
}

void receipt(string* receiptMsg, string* payMethod) {
        char confirm;
        string seatRow, seatCol;
        cout << "\n\n";
        cout << setw(66) << "Print receipt? (Y/N): ";  //confirmation to print receipt
        cin >> confirm;
        while (toupper(confirm) != 'Y' && toupper(confirm) != 'N') {
                cout << setw(66) << "Invalid input. Only Y/N: ";
                cin >> confirm;
        };
        if (toupper(confirm) == 'Y') {
                system("CLS");
                //receipt variable: category, amount, seats, total price, payment method,
invoice no., receiptMsg
                cout << "\n\n";
                cout << "\t
_____" << endl;
                cout << "\t| " << setw(64) << "|" << endl;
                cout << "\t| " << setw(53) << "+---------------------------------------+" << setw(11)
<< "|" << endl;
                cout << "\t| " << setw(53) << "|    _____         __  ___  __          |" << setw(11)
<< "|" << endl;
```

```
cout << "\t| " << setw(53) << "|   / ___/__  ___  / / / _ \\/ /__ ___ __  |" <<
setw(11) << "|" << endl;
cout << "\t| " << setw(53) << "|  / /__/ _ \\/ _ \\/ / ___/ / _ `/ // /  |" << setw(11)
<< "|" << endl;
cout << "\t| " << setw(53) << "|  \\___/\\___/\\___/_/ /_/  /_/\\_,_/\\_, /   |" <<
setw(11) << "|" << endl;
cout << "\t| " << setw(53) << "|                                /___/    |" << setw(11) << "|"
<< endl;
cout << "\t| " << setw(53) << "+----------------------------------------+" << setw(11)
<< "|" << endl;
cout << "\t| " << setw(38) << " ** RECEIPT ** " << setw(26) << "|" << endl;
cout << "\t| " << setw(64) << "|" << endl;
cout << "\t| " << "\t\tInvoice number\t: INV000" << cust_num + 1 << setw(25)
<< "|" << endl;
cout << "\t| " << "\t\tDate\t\t: 17/12/2023" << setw(22) << "|" << endl;
cout << "\t|   " <<
"============================================================" << setw(4) <<
"|" << endl;
cout << "\t| " << setw(64) << "|" << endl;
cout << "\t|   " << setw(20) << left << "Ticket category" << setw(10) << "Qty"
<< setw(15) << "Price (RM)"
        << setw(15) << "Amount" << right << setw(2) << "|" << endl;
if (currentVIPAmount != 0) {
        cout << "\t|   " << setw(20) << left << ticketType[0] << setw(10) <<
currentVIPAmount << setw(15) << ticketPrice[0]
                << setw(15) << currentVIPAmount * ticketPrice[0] << right <<
setw(2) << "|" << endl;
}
if (currentGenAmount != 0) {
        cout << "\t|   " << setw(20) << left << ticketType[1] << setw(10) <<
currentGenAmount << setw(15) << ticketPrice[1]
                << setw(15) << currentGenAmount * ticketPrice[1] << right <<
setw(2) << "|" << endl;
}
cout << "\t| " << setw(64) << "|" << endl;
cout << "\t|\t\t" << "Seat(s) code\t: " << setw(32) << "|" << endl;
for (int i = 0; i < VIPROW; ++i) {
        for (int j = 1; j <= VIPCOL; ++j) {                    //output all seats
currently chosen
                if (currentVIPSeats[i][j - 1]) {
                        seatRow = "V" + string(1, char(i + 'A'));        //add 'V' in
front of seat code to indicate vip seats
                        seatCol = to_string(j);
                        cout << "\t| " << "\t\t\t\t  " << setw(20) << left <<
seatRow + seatCol << right << setw(12) << "|" << endl;
                };
        };
};
```

```
                for (int i = 0; i < GENROW; ++i) {
                        for (int j = 1; j <= GENCOL; ++j) {
                                if (currentGenSeats[i][j - 1]) {
                                        seatRow = string(1, char(i + 'A'));
                                        seatCol = to_string(j);
                                        cout << "\t| " << "\t\t\t\t  " << setw(20) << left <<
seatRow + seatCol << right << setw(12) << "|" << endl;
                                };
                        };
                };
                cout << "\t| " << setw(64) << "|" << endl;
                cout << "\t|   " << "-------------------------------------------------------" << setw(4) <<
"|" << endl;
                cout << "\t| " << "\t\tTotal Price\t: RM " << setw(20) << left << fixed <<
setprecision(2) << currentTotal << right << setw(9) << "|" << endl;
                cout << "\t| " << "\t\tPayment Method\t: " << setw(20) << left << *payMethod
<< right << setw(12) << "|" << endl;
                cout << "\t| " << "\t\t" << setw(40) << left << *receiptMsg << right << setw(10)
<< "|" << endl;
                cout << "\t| " << setw(64) << "|" << endl;
                cout << "\t ------------------------------------------------------------" << endl;
                cout << "\n\n";
                cout << setw(50) << "Please take the receipt." << endl;
        };
        *receiptMsg = "";
        *payMethod = "";
}

void ticket() {
        string seatRow, seatCol;
        char next;
        int queue = tickets_sold + 1;
        cout << "\n\n";
        cout << setw(100) << "----------------------------------------" << endl;
        cout << setw(86) << "Tickets Generated" << endl;
        cout << setw(100) << "----------------------------------------" << endl << "\n\n";

        //ticket variable: category, price, seat, queue no.
        for (int i = 0; i < VIPROW; ++i) {
                for (int j = 1; j <= VIPCOL; ++j) {
                        if (currentVIPSeats[i][j - 1]) {   //if current seats chosen (true), print
ticket for that seat
                                seatRow = "V" + string(1, char(i + 'A'));        //add 'V' in front of
seat code to indicate vip seats
                                seatCol = to_string(j);
                                queue -= 1;
                                cout << "\n\n";
```

```
                                        cout << "\t\t
_____________________________________________________________________________________________________________" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << "
*VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**
VIP**VIP* |" << endl;
                                        cout << "\t\t| " << setw(10) << left << ticketType[0] << right <<
"\t\t|  " << setw(70) << "|" << " Venue :" << setw(18) << " |" << endl;
                                        cout << "\t\t| RM\t" << fixed << setprecision(2) << ticketPrice[0]
<< "\t\t|  " << "   \\  |  |  |   __| _ _|   __|           _|    |    |         " << setw(8) << " |" << setw(26) <<
" |" << endl;
                                        cout << "\t\t|" << "\t\t\t| " << "   |\\/ |  |  | \\__ \\   |   (        _ \\   _|
_|    \\    -_)" << setw(8) << "|" << " National Stadium" << setw(9) << " |" << endl;
                                        cout << "\t\t| **FREE T-SHIRT" << "\t| " << "  _|__|| \\_\\/|
_|__/_|__| \\___||  \\__|/ _|     \\__| _| _| \\___| " << setw(8) << " | " << "Bukit Jalil" << setw(15)
<< "  | " << endl;
                                        cout << "\t\t| **FREE GLOW STICK" << "\t| " << "  \\__ \\  __ /
__ | _|     /  _ | \\__ \\  " << setw(32) << " | " << "Kuala Lumpur" << setw(14) << "  | " << endl;
                                        cout << "\t\t|" << "\t\t\t| " << "  ____/ _|  _| _| ___| _|_\\ ___|
____/" << setw(33) << "|" << setw(26) << " |" << endl;
                                        cout << "\t\t| GATE\t" << "15A" << "\t\t| " << setw(71) << "|" <<
setw(26) << " |" << endl;
                                        cout << "\t\t| SEAT\t" << seatRow << seatCol << "\t\t| " <<
setw(50) << ">> COOLPLAY WORLD TOUR <<" << setw(21) << "|" << " Wednesday" <<
setw(16) << " |" << endl;
                                        cout << "\t\t|" << "\t\t\t| " << setw(55) << " ______
_____ " << setw(16) << "|" << " December 22, 2023" << setw(8) << " |" << endl;
                                        cout << "\t\t|" << "\t\t\t| " << setw(55) << " ( ___
)--------------------( ___ )" << setw(16) << "|" << setw(26) << " |" << endl;
                                        cout << "\t\t| GATE Open 6.30pm " << "\t| " << setw(56) << "  |
|  _   _   ___   ____   |   |  " << setw(15) << "|" << "    ____________________" << setw(4) << "
|" << endl;
                                        cout << "\t\t|" << "\t\t\t| " << setw(54) << "|   | \\ ( ) / )_ _( )  _)\\  |
|" << setw(17) << "|" << "   | \" All MUSIC" << setw(7) << "|" << setw(3) << " |" << endl;
                                        cout << "\t\t|" << "\t\t\t| " << setw(54) << "|   |  )\\_/(  _| |_ | '__/  |
|" << setw(17) << "|" << "   |   IS BEAUTIFUL \" | " << setw(2) << " |" << endl;
                                        cout << "\t\t| Queue No. " << queue << "\t\t| " << setw(54) << "|
|   \\_/  )_____()_(     |   |" << setw(17) << "|" << "    ------------------" << setw(4) << " |" << endl;
                                        cout << "\t\t|" << "\t\t\t| " << setw(54) << "|___|
|___|" << setw(17) << "|" << setw(26) << " |" << endl;
                                        cout << "\t\t|" << "\t\t\t| " << setw(55) <<
"(_____)--------------------(_____)" << setw(16) << "|" << " C  O  O  L  P  L  A  Y" << "  |" <<
endl;
                                        cout << "\t\t|" << "\t\t\t|" << "
*VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**VIP**
VIP**VIP* |" << endl;
```

```
                                        cout << "\t\t
-----------------------------------------------------------------------------------------------------------------------"
<< endl;

                                }
                        }
                }
        for (int i = 0; i < GENROW; ++i) {
                        for (int j = 1; j <= GENCOL; ++j) {
                                if (currentGenSeats[i][j - 1]) {
                                        seatRow = string(1, char(i + 'A'));
                                        seatCol = to_string(j);
                                        queue -= 1;
                                        cout << "\n\n";
                                        cout << "\t\t
_____________________________________________________________________________________________________________________" << endl;
                                        cout << "\t\t| " << setw(10) << left << ticketType[1] << right <<
"\t\t| " << setw(45) << " **COOLPLAY WORLD TOUR** " << setw(25) << "|" << " Venue :" <<
setw(18) << "|" << endl;
                                        cout << "\t\t| RM\t" << fixed << setprecision(2) << ticketPrice[1]
<< "\t\t| " << "   __  __            _                __   _   _                 " << setw(2) << "|" << setw(26)
<< "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << "   |  \\/  |           (_)              / _| | |  |
|           " << setw(5) << "|" << " National Stadium" << setw(9) << "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << "   | \\  / | _   _ ___  _   ___    ___  |
|_  | |_ | |__    ___   " << setw(5) << "|" << " Bukit Jalil" << setw(14) << "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << "   | |\\/| || | | |/ __|| | / __|  / _ \\ |  _| |
__|| '_ \\  / _ \\  " << setw(5) << "|" << " Kuala Lumpur" << setw(13) << "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << "   | |  | || |_| |\\__ \\| || (__  | (_) || |   |
|_ | | | ||  __/" << setw(7) << "|" << setw(26) << "|" << endl;
                                        cout << "\t\t| GATE\t" << "15A" << "\t\t|" << "   |_|  |_|
\\__,_||___/|_| \\___|  \\___/ |_|    \\__||_| |_| \\___|" << setw(7) << "|" << setw(26) << "|" <<
endl;
                                        cout << "\t\t| SEAT\t" << seatRow << seatCol << "\t\t|" <<
setw(71) << "|" << " Wednesday" << setw(16) << "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << setw(55) << " _____         _
" << setw(16) << "|" << " December 22, 2023" << setw(8) << "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << setw(55) << " / ____|       | |
" << setw(16) << "|" << setw(26) << "|" << endl;
                                        cout << "\t\t| GATE Open 7.00pm " << "\t|" << setw(55) << " |
(___   _ __  | |__    ___  _ __  ___  ___ " << setw(16) << "|" << "    ____________________" <<
setw(4) << "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << setw(55) << "  \\___ \\ | '_ \\ | '_ \\  / _
\\| '__|/ _ \\/ __|" << setw(16) << "|" << "   | \" WHERE MEMORIES | " << setw(2) << "|" <<
endl;
                                        cout << "\t\t|" << "\t\t\t|" << setw(55) << "  ____) || |_) || | | ||  __/|
|  |  __/\\__ \\" << setw(16) << "|" << "   |    ARE CREATED \" | " << setw(2) << "|" << endl;
```

```
                                        cout << "\t\t| Queue No. " << queue << "\t\t|" << setw(55) << "
|_____/ | .__/ |_| |_| \\___||_|   \\___||___/" << setw(16) << "|" << "    ------------------" << setw(4)
<< "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << setw(54) << "        | |
" << setw(17) << "|" << setw(26) << "|" << endl;
                                        cout << "\t\t|" << "\t\t\t|" << setw(55) << "         |_|
" << setw(16) << "|" << " C  O  O  L  P  L  A  Y" << "  |" << endl;
                                        cout << "\t\t
-------------------------------------------------------------------------------------------------------------------------"
<< endl;
                                };
                        };
                };
        cout << "\n\n\n";
        cout << setw(75) << "Press [Y] if done ... ...";
        cin >> next;
        while (toupper(next) != 'Y') {
                cout << setw(70) << "Enter [Y] only: ";
                cin >> next;
        };
}

void salesReport() {
        char next;
        int vipSold = VIPROW * VIPCOL - vipAvailib;
        int genSold = GENROW * GENCOL - generalAvailib;
        double vipProfit = vipSold * ticketPrice[0];
        double genProfit = genSold * ticketPrice[1];

        cout << setw(110) <<
"=========================================================================
==============" << endl << endl;
        cout << setw(97) << " ____        _            ____                    _   " << endl;
        cout << setw(97) << "/ ___|  __ _| | ___  ___  |  _ \\ ___ _ __   ___  _ __| |_ " << endl;
        cout << setw(97) << "\\___ \\ / _` | |/ _ \\/ __| | |_) / _ \\ '_ \\ / _ \\| '__| __|" << endl;
        cout << setw(97) << " ___) | (_| | |  __/\\__ \\ |  _ <  __/ |_) | (_) | |  | |_ " << endl;
        cout << setw(97) << "|____/ \\__,_|_|\\___||___/ |_| \\_\\___| .__/ \\___/|_|   \\__|" <<
endl;
        cout << setw(97) << "                                    |_|                  " << endl;
        cout << setw(60) << "Date: 17 December 2023" << endl;
        cout << setw(61) << "Event: CoolPlay Concert" << endl;
        cout << setw(57) << "Sales Person: user1" << endl;
        cout << setw(110) << "-------------------------------------------------------------------------------"
<< endl;
        cout << "\n";

        cout << setw(90) << "+=========================================+" << endl;
        cout << setw(90) << "|____  ___________  ___  ________  ________|" << endl;
```

```
	cout << setw(90) << "||  |  |__|__|__|\\ ||   [__|__|  |__[__ |" << endl;
	cout << setw(90) << "||__|__| \\| \\|__| \\||   ___|  |__|_____]|" << endl;
	cout << setw(90) << "+=========================================+" << endl
<< endl;
	cout << setw(108) << "   TypeSeat      | PRICE (RM) | TOTAL SEAT | SEAT LEFT |
SEAT SOLD | PROFIT (RM)" << endl;
	cout << setw(108) << "-----------------|------------|------------|-----------|-----------|------------"
<< endl;
	cout << setw(72) << "      VIP        |     788.90 |         30 |" << setw(11) << vipAvailib <<
"|" << setw(11) << vipSold << "|" << setw(11) << fixed << setprecision(2) << vipProfit << endl;
	cout << setw(72) << "    GENERAL      |     250.90 |        140 |" << setw(11) <<
generalAvailib << "|" << setw(11) << genSold << "|" << setw(11) << fixed << setprecision(2)
<< genProfit << endl;
	cout << setw(108) << "-------------------------------------------------------------------------------"
<< endl;
	cout << setw(46) << " TOTAL PROFIT(RM)" << setw(61) << vipProfit + genProfit <<
endl;
	cout << setw(108) << "-------------------------------------------------------------------------------"
<< endl;
	cout << setw(52) << "Number of customer(s): " << cust_num << endl;
	cout << "\n";

	int data[2] = { vipSold, genSold }; //data for number of dots to be display (1 dot = 2
tickets)
	string higher = ticketType[0];
	string lower = ticketType[1];
	if (genSold > vipSold) {
		data[0] = genSold;
		data[1] = vipSold; //find highest data for bar chart
		higher = ticketType[1]; //let higher bar chart to be displayed first
		lower = ticketType[0];
	}
	for (int dot = data[0]; dot > 0; dot--) { //number of dots for highest bar chart
		cout << "\t\t\t\t\t";
		for (int bar = 0; bar < 2; bar++) { //number of bar for 2 categories
			if (data[bar] >= dot) {
				cout << setw(17) << "*";
			}
			else {
				cout << " ";
			}
		}
		cout << endl;
	}
	cout << "\t\t\t\t\t" << setw(18) << higher << setw(18) << lower << endl << endl;
	cout << setw(111) <<
"=================================================================
==============" << endl;
```

```
	cout << setw(55) << "Note that one * = 1 tickets" << endl;
	cout << "\n\n\n";
	cout << setw(75) << "Press [Y] back to menu : ";
	cin >> next;
	while (toupper(next) != 'Y') {
		cout << setw(70) << "Enter [Y] only: ";
		cin >> next;
	};
}
```